

The role of strategic institutional ownership in the decision of emerging market leadership: Evidence from Chinese cross-border listings

Fan “John” Zhang*

(First Version: August 2018)
(This Version: July 2020)

ABSTRACT

This paper investigates the influence of strategic institutional ownership on the leadership decision of emerging market cross-border listings. Using a sample of Hong Kong-listed firms from the Chinese mainland, it shows that institutional ownership is positively related to the probability of cross-border listings. The effect holds for different listing types, after addressing endogeneity issues, and with implementing exogenous shocks. The evidence also suggests that while political considerations also influence cross-border listings, it does not weaken the overall effect of institutional ownership. Further analyses reveal that the positive effect of institutional ownership on cross-border listings is associated with informational environment improvement and that small firms located in better social developed regions are more likely to attract institutional investors. Overall, this study provides evidence that institutional ownership helps cross-border listings, which in turn benefits emerging market firms.

Keywords: Emerging market leadership; institutional investor; cross-listing; ownership structure; China

JEL classification: G23, G15, G32, O19, R11

* The paper benefits from valuable feedback and comments from Renée B Adams (Oxford University), Christina Ahmadjian (Hitotsubashi University), Narjess Boubakri (American University of Sharjah), Jens Christensen (Federal Reserve Bank of San Francisco), Elizabeth Demers (University of Lausanne), Stav Fainshmidt (Florida International University), Georg Schaur (University of Tennessee), Lemma Senbet (University of Maryland), Guochang Zhang (Hong Kong University of Science and Technology) and seminar participants at the 94th Western Economic Association International (WEAI) Annual Conference and the 2019 International Banking, Economics, and Finance Association (IBEFA) Summer Meeting. All errors are the author’s responsibility. Please communicate by contacting fax: (853) 2882-3281 or postal address: Avenida Wai Long, the School of Business, Macau University of Science and Technology Taipa, Macau. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

1 Introduction

One of the important topics in current international financial economic research is how to integrate emerging markets with the world economy (Bekaert, 1995). One way to achieve this purpose is to encourage emerging market firms to take part in financial activities in the developed economies, such as issuing shares in developed markets (Edison & Warnock, 2004; Karolyi, 2004, 2012; Lins, 2003).¹ However, in emerging economies, corporations are usually controlled by insiders who would have stronger incentives to withhold value-relevant private information (Claessens, Djankov, & Lang, 2000; Lang, 2005).² On the other hand, the exchanges in the developed market require high disclosure standards and strict rules, this would limit the controlling shareholder, the largest shareholder, and insiders to consume their private benefits (Doidge, Karolyi, Lins, Miller, & Stulz, 2009). As a result, many emerging market firms are reluctant to list overseas, even knowing cross-border listings can bring firms with high potential benefits.

This study argues that institutional ownership helps the cross-border listing of emerging market firms. Literature has documented that institutional investors actively support corporate initial public offerings (IPOs) and play a positive role in the improvement of corporate information environment (Chemmanur & He, 2011; Chemmanur, He, & Hu, 2009; Chemmanur, Hu, & Huang, 2010; Lins, 2003), though the original motivation for institutional investors participating in the IPO activities is that they can reap considerable profits.³ The informational role of institutional investors is crucial for emerging market

¹ The benefits of the cross-border listing include, for example, an increased market liquidity, an easier way to access capital, a reduced agency conflict by bonding to the high legal requirements of the listing destinations, an improved information environment, an efficient monitoring by capital markets, and a diversification channel for investors in developed markets (Karolyi, 2004, 2006, 2012). In terms of corporate governance, listing overseas can improve governance standard and quality by raising both direct constraints (such as stricter law and disclosure requirements) and indirect constraints (such as increased analyst scrutiny) (Doidge et al., 2009). See Literature Review Section for a detailed discussion of the benefits of cross-border listings of emerging market firms.

² Jiang and Kim (2015) document that the largest shareholders in China hold substantial equity ownership in listed firms (on average more than 40% as of 2014) and that a manager can become entrenched if the manager is also a large controlling shareholder. Doidge et al. (2009) document that firms with controlling shareholders, with the largest shareholder's control rights exceeding cash flow rights, and with insiders (such as family and management members) assuming control are less likely to list to the U.S. stock exchanges.

³ For example, Jenkinson and Jones (2008) find that institutions do have a significant influence on the IPO allocations (i.e. how investors assess IPOs). Chiang et al. (2009) show that institutional investors are generally bid a fair price in IPOs relative to individual investors, suggesting institutional investors are better informed. The information advantage of institutional investors in stock listings has been further documented in the

firms because, with improved information transparency, emerging market firms are more likely to have better corporate governance and thus confirm the rules of developed stock exchanges. Further institutional investors are an important countervailing power to corporate insiders (Farrar & Girton, 1981). Consistent with the prior literature, this paper defines the institutional ownership as the strategic institutional ownership whose holdings are not for speculative tradings (thus, the holdings should have material influence on long-term decisions such as cross-border listing), and whose holdings are not restricted (thus, the institutional shares should be free float shares, with which institutional investors can buy or sell freely). Therefore, if an emerging market firm has institutional shareholders in their ownership structure, it would be expected an increased probability of cross-border listing of the firm.

To examine whether institutional ownership facilitates the cross-border listing of emerging market firms, this study collects a sample of firms from the Chinese mainland⁴ and listed on the Hong Kong stock exchange (HKSE) over the period from 1990 to 2017. China is an excellent platform to investigate the effect of stock listings from the emerging economy to the developed exchange. It is well understood that in the Chinese mainland, quotas of initial public offerings (IPOs) are distributed by the government and political considerations play an important role in such quota distribution (see, for example, Hung, Wong, & Zhang, 2012; Sun, Tong, & Wu, 2013). More than 60% of firms listed on the Shanghai and Shenzhen Stock Exchanges are state-owned enterprises (SOEs). These quotas limit the possibility of domestic listings, Chinese firms thus view cross-border listings as an important alternative and a more feasible way to issue stocks to the public. Nonetheless, the unique characteristics in China also have a significant influence on firms' cross-border listing decisions. According to Hung et al. (2012), politically connected firms are more likely to list overseas than non-politically connected firms and post-overseas listing's performance is worse for the connected firms compared to the non-connected firms, suggesting that managers of politically connected firms list their firms overseas for their political benefits. Given the circumstance of significant political influences on listing decisions, there is a need

literature on seasoned equity offerings (SEOs). For example, Chemmanur et al. (2009) report that institutional investors do possess private information in SEOs and can obtain benefits from trading such information.

⁴ The Chinese mainland is also known as Mainland China, which does not include the special administrative regions of Hong Kong and Macau, and Taiwan.

for solutions for private Chinese firms. If institutional ownership is one of such solutions as the driver of cross-border listings and generating positive results for private Chinese firms, then the effect of institutional ownership is expected to be even stronger in countries with less government influence.

Further, the HKSE provides a complete list of Chinese firms that issue shares in Hong Kong and comprehensively disclose the types of listings. This rich data set allows looking into such sophisticated cross-border listing activities in the emerging markets as the ones in China. Moreover, the HKSE is an advanced stock exchange that provides more transparent information regarding stock listings. What is more, there is a low cultural difference between Hong Kong and China. Studies have shown that country-level proximity is an important determinant of cross-border listing decisions (Sarkissian & Schill, 2004). By focusing on the Hong Kong stock market, the influence of cultural differences on cross-border listing decisions is minimized. Last but not least, Chinese firms have been actively participating in international listing activities. With a sufficient sample, Hong Kong-listed Chinese stocks offer an excellent channel to investigate emerging market cross-border listings.

In line with the expectation, the result shows a positive relation between institutional ownership and the likelihood of all types of Chinese listings on the HKSE. In particular, the positive relation is stronger for the firms that incorporated in the Chinese mainland (so-called H-shares) and for the firms that are listed on the mainboard of the HKSE.⁵ These results provide supporting evidence that institutional ownership helps promote the cross-border listing of emerging market firms. To ensure the robustness of the main results, the paper addresses several endogeneity issues. First, to mitigate potential reverse causality that cross-border listings may cause institutional ownership to increase, the two-stage least squares (2SLS) regressions are employed. The results show that the positive relation between institutional ownership and the possibility of cross-border listings does not alter. This suggests that the relation found previously can be interpreted causally, that is, the greater the institutional ownership of

⁵ A-share firms are both incorporated and listed in the Chinese mainland, H-share firms are incorporated in the Chinese mainland but listed in Hong Kong, Red-chip firms are incorporated overseas and listed in Hong Kong, and AH-share firms are incorporated in the Chinese mainland and listed in both the Chinese mainland and Hong-Kong markets. See https://www.hkex.com.hk/Market-Data/Statistics/Consolidated-Reports/China-Dimension?sc_lang=en.

a Chinese firm, the higher the likelihood of cross-border listing of the firm. To alleviate this effect is not a result of omitted variables, the propensity score-match is used. With four different matching techniques, institutional ownership is consistently and positively associated with cross-border listings.

To further check the robustness, a difference-in-differences (DID) test is performed with the implement of exogenous structure change of the Chinese market. The Chinese government has continuously reformed its IPO system. In earlier years, the issuers in the mainland market may have no incentive to invite strategic institutional investors, and there are not many strategic institutional investors in the mainland capital market. However, since Chinese policymakers made regulations to reinforce institutional investors into the capital market and considered cross-border listing as an important measure to reform the ownership structure of SOEs (Hung et al., 2012), it is likely to observe more strategic institutional investors for firms listed in HKSE than those listed in the mainland market. Thus, the role of institutional ownership may be more valuable for cross-border-listing in the latter stage. Further, the Chinese domestic market demand for IPO is always high, which guarantees the success of almost any IPO with the quota. However, the issuers from the mainland cannot guarantee their success of IPO in the Hong Kong Stock Exchange and thus they may invite strategic investors to join the IPO. Our result suggests that there is a consistently strong effect on the interaction between institutional ownership with each stage of structure change of the Chinese market. As the institutional ownership without interactions remains strong in these regressions, the association between the institutional ownership and the cross-border listing is not simply an outcome from the voids in the market institutions, implying the promoting effect for cross-border listing by institutional investors. In fact, the results show that the effect of institutional ownership not only withstands the exogenous shocks, but it is also enhanced by these shocks. After addressing endogeneity issues, these tests confirm the robustness of the main finding that institutional ownership plays a significant role in cross-border listings.

Next, we take into account key institutions in China market. Especially, we investigate whether the main results are affected by political considerations. Several approaches are adopted to address this issue. The first approach is to focus on Beijing-based firms, Beijing is the capital city of China and more likely to subject to political influences. If political considerations overwhelm the effect of cross-border

listings, then the effect of institutional investors may not function well in Beijing-based firms. While the result confirms the political influences of Beijing-based firms on their cross-border listings, the presence of institutional investors remains a significant and positive effect on the probability of cross-border listings. In the regions outside of Beijing, institutional ownership consistently shows a significant effect on cross-border listings in both developed and less developed regions. Therefore, the role of institutional investors in promoting cross-border listings is neither totally eroded by political influences nor geographically dependent.

Further, the paper focuses on a subsample of firms with foreign institutional ownership. It is fairly unlikely that foreign institutions are controlled by the Chinese government. In contrast, at least some of the domestic institutional investors, such as banks and insurance companies, may not be politically independent. Therefore, an examination of the effect foreign institutional ownership can alleviate the concern that the effect of institutional shareholdings on Chinese cross-listings is driven by government decisions. Besides, looking at foreign institutions can also address the issue that cultural proximity may affect cross-border listing decisions (Bena, Ferreira, Matos, & Pires, 2017; Ferreira & Matos, 2008). As one may argue that Hong Kong is culturally close to China, and therefore, the main result is driven by domestic institutional investors who may be more likely to choose Hong Kong as the listing destination. The subsample test shows that emerging market firms with a higher proportion of foreign institutional investors have a stronger explanatory power for cross-border listings, compared to the whole sample that includes both domestic and foreign institutional investors. This does not support the argument that the role of institutional investors in cross-border listings is driven by the government or purely by cultural similarity associated with domestic institutional investors. Therefore, the evidence here is in line with the literature that emphasizes that foreign institutional ownership disciplines managerial entrenchment (Bena et al., 2017; Dahlquist & Robertsson, 2001), which is more consistent with the requirements of developed exchanges.

If the logic behind the main result that institutional investors promote cross-border listings is that they emphasize better corporate governance, then it would be expected that institutional ownership mitigates governance issues in emerging market firms through stock listings. Prior literature has documented that

institutional ownership in emerging market firms is closely related to the effectiveness of corporate governance. For example, Chemmanur and Liu (2011) and Lins (2003) show that institutional investors contribute to the improvement of managerial efficiency by mitigating agency costs and information asymmetry. On the other hand, Gibson (2003) finds that firms with a large number of domestic individual shareholders are more likely to have inefficient corporate governance. Further, Lins (2003) reports that introducing outside non-managerial institutional ownership is associated with firm valuation and ownership structure improvements for emerging market firms. These findings suggest that institutional ownership can influence the corporate governance and strategic decision-making of emerging market firms in a positive way.⁶

To uncover whether better corporate governance is the underlying logic that institutional investors promoting the cross-border listing of emerging market firms, this paper explicitly tests their effects on the corporate information environment and government control that have been emphasized in the literature. The results show that institutional ownership helps the improvement of the information environment and constrain government ownership. These results are in line with the literature and provide evidence supporting that by promoting cross-border listings, institutional holdings benefit the governance of emerging market firms. Finally, to seek the explanation of institutional ownership, this paper follows standard ownership structure and cross-border listing literature (such as Ayyagari & Doidge, 2010; Doidge et al., 2009) and employ Cox's (1972) proportional hazard model. The result suggests that institutional investors are more likely to have strategic holdings in small firms located in regions where the level of social development is close to Hong Kong.

By focusing on the Chinese market, the current largest emerging economy in the world, this study sheds light on the question that how to integrate emerging markets into the world's economy. As pointed out by Bekaert (1995), market integration is essential for investors to generate returns globally. While prior

⁶ This is in particular important for emerging market firms whose initial governance and strategy are usually problematic. For example, Masayoshi Son, CEO of Softbank, the largest institutional shareholder of Alibaba once quotes, "When Jack Ma came to me, he had no business plan with zero revenue. His business model was wrong." (<https://inshorts.com/en/news/jack-ma-had-no-business-plan-and-zero-revenue-softbank-ceo-1507817627899>).

economic studies, such as Edison & Warnock (2004) and Karolyi (2004), affirm that the importance of cross-border listings in this regard, the underdeveloped emerging market corporate governance remains a significant hindrance to the cross-border listings (Karolyi, 2012). Especially, the ownership structure problems are pervasively observed in emerging markets. For example, larger shareholders and insiders tend to expropriate minority shareholders and the government controls a firm to assume political purposes. Realizing these problems, Chinese policymakers have made regulations to reinforce institutional investors into the capital market since 2004 and considered cross-border listing as an important measure to reform the ownership structure of SOEs (Hung et al., 2012). However, the effectiveness of these measures has not been empirically explored in the literature. The results of this paper disclose that institutional ownership is positively related to the propensity of the cross-border listings, through which they further play a prominent role in the optimization of the information environment and the reform of government ownership of emerging market firms.

This paper is particularly related to Ferreira and Matos (2008). Both this paper and theirs show that institutional ownership benefits companies. Ferreira and Matos (2008) argue that institutional investors monitor corporate governance worldwide and find that firms with large institutional holdings are characterized by higher valuations, better operating performance, and lower capital expenditures. This paper extends Ferreira and Matos (2008) in the following main aspects. First, Ferreira and Matos (2008) offer evidence that institutional investors are more likely to invest in firms that are already listed in the developed stock exchanges, whereas this paper investigates the effect of institutional ownership on cross-border listing decision-makings. Therefore, the results in this paper and theirs together demonstrate that institutional investors are important to a firm in both pre- and post-listing periods. Second, Ferreira and Matos (2008) focus on foreign institutional investors, whereas this paper includes both local and foreign institutional investors (though it does not separately test and confirm the effect of foreign institutional investors). Therefore, this paper provides support for the findings documented in the existing studies that both local and foreign institutional ownership helps reduce information asymmetry (Baik, Kang, & Kim, 2010; Bena et al., 2017). Third, Ferreira and Matos (2008) carry out an analysis by focusing on the exchange side (U.S. stock exchanges), whereas this firm focuses on the

firm side (emerging market firms). The backgrounds of foreign firms listed on a developed exchange can be highly heterogeneous, as they represent firms from both emerging and developed markets. By focusing on the Chinese firms, this study alleviates this heterogeneity issue and intensively addresses cross-border listing issues in an emerging market.

This paper also contributes to the literature on foreign listings and cross-listings. As studies have shown, the ownership structure significantly influences overseas listing decisions.⁷ This paper connects to this literature and offers evidence that overseas listings improve the quality of corporate governance of emerging market firms. To this end, this study has some practical significances. On the one hand, the problematic ownership structure in the emerging market is a cause of suboptimal corporate governance. On the other hand, both the presence of institutional ownership and cross-border listings are helpful for the improvement of corporate governance.⁸ This paper shows that institutional ownership facilitates the cross-border listing, which is important for emerging market firms to optimize their corporate governance and facilitate their integration to the global financial market.

The remainder of the paper proceeds as follows. Section 2 reviews the literature that discusses the strategic role of institutional investors. Section 3 describes the data, sample, and variables. Section 4 reports empirical results. Section 5 concludes the paper.

2 Literature review

This section reviews the literature on the cross-border listing and the strategic role of institutional investors. It first overviews the potential benefits for emerging market firms to carry out cross-border listings. Then, it discusses the challenges that can impede cross-border listings of emerging market

⁷ For example, Doidge et al. (2009) find that when blockholders' control rights exceed their cash flow rights and when insiders (such as family members or managers) are the controlling shareholders, the possibility of listing on a U.S. major exchange significantly decreases. They argue that this is because the U.S. market enforces stringent rules and requires higher levels of information disclosure, which goes against insiders' private benefits. Further, Ayyagari and Doidge (2010) show that after listing on the exchange in the developed market such as the United States, controlling shareholders of a foreign firm are likely to lose the voting rights, suggesting that cross-border listing optimizes corporate ownership and control.

⁸ However, this does not mean that the institutional investor can be a substitution for the cross-border listing. The cross-border listing and introducing institutional investors are two different corporate decision-makings. While we find that introducing institutional investors is helpful for the firm to list overseas, our results do not suggest that the function of cross-border listings can be replaced by the function of introducing institutional investors.

firms. Finally, it points out how institutional investors help overcome these obstacles.

2.1 The cross-border listing of emerging market firms

2.1.1 Benefits

For emerging market firms, the direct benefit of the cross-border listing on the developed market lies in better legal institutions. Firms in emerging market economies that generally have a lower governance standard and weaker investor protection. According to the bonding hypothesis, listing on a developed market enables these firms to overcome country-level institutional limitations and increase firm-level governance quality (Doidge, Karolyi, & Stulz, 2007; Karolyi, 2012). By listing on exchanges in developed markets, controlling shareholders and firm insiders “bond” themselves to the higher requirements of these exchanges and forsake unreasonable private benefits (Karolyi, 2012). Conceptually, the “bonding” mechanisms can be classified as “legal bonding” and “reputational bonding”. The former stresses the regulatory enforcement and the latter emphasizes the role of “reputational” intermediaries or capital market participants, such as analysts, media, investment bankers, auditors, and lawyers (Karolyi, 2012). As empirical evidence, Fernandes and Ferreira (2008) find that cross-border listing is associated with increased analyst coverage and thus increased information production. This has strong explanatory power for cross-border listings of emerging market firms but not for the cross-listings of firms from developed markets.

Further, the literature documents that cross-border listings offers materialized benefits for the listed firms. This is because the developed stock market provides a better valuation mechanism (Doidge, Karolyi, & Stulz, 2004; King & Segal, 2009; Sundaram & Logue, 1996). Karolyi (2006) argues that one benefit of listing overseas is to ensure access to the external capital market in the future. Indeed, Lins et al. (2005) document that emerging market firms gain greater access to external capital after listing in the U.S. stock market. Lins et al. (2005) argue that cross-border listings significantly reduce the indirect barriers for emerging market firms. As empirical results, they find that emerging market firms raise capital more frequently than developed market firms and do not report financial constraints after cross-border listing.

Cross-border listings also benefit the overall stock market development for both emerging and developed economies. In terms of emerging economies, Karolyi (2004) examines the impact of American depository receipts (ADRs) of U.S. listed emerging market firms on the development of their stock markets and finds that ADRs foster greater cross-border capital flows, higher market capitalization, larger numbers of listed firms, and more share trading volumes.⁹ In terms of developed markets, cross-border listed emerging market firms also provide important diversification channels for investors in developed markets. Edison and Warnock (2004) study U.S. investors' equity positions in emerging markets, finding that U.S. investors have a strong preference for stocks that are listed on U.S. exchanges and that the weight of these emerging market firms in U.S. portfolios are in line with the prediction by a float-adjusted *International Capital Asset Pricing Model (ICAPM)*, suggesting that investors will integrate U.S.-listed emerging market firms into their U.S. portfolios.¹⁰

2.1.2 Challenges

While cross-border listings bring benefits to both emerging and developed markets, there are still serious challenges that prevent them from being carried out easily. The first challenge is the capital market segmentation. Stulz (2005) finds that although explicit barriers to international investments have been significantly removed over the last decades, financial markets between countries remain segmented. One cause of such segmentation is that the interests between corporate insiders and outside investors cannot always be aligned. To assume control, insiders tend to retain substantial equity and limit globalization. The alignment of their interests becomes even less likely when corporate insiders and outside investors reside in different countries. This is particularly problematic given that after the cross-border listings, the managers remain in the emerging market and investors are in the developed market (Bekaert, 1995). Moreover, the financial segmentations between countries are formed through

⁹ However, these benefits do not accrue to those firms of the non-ADR segment. The explanation for these results is that non-ADR programs do not associate with greater stock trading activities, broader ownership, and more frequent capital-raising activities as ADR programs do. Given that cross-border listing provides emerging market firms with a high-quality information environment and platform for capital raising, some scholars concern that the increasing trend of globalization would intensify the migration of stock listing conducted by emerging market firms to the developed exchanges, putting their domestic stock market under a greater pressure (Claessens, Klingebiel, & Schmukler, 2002; Domowitz, Glen, & Madhavan, 1998).

¹⁰ The authors argue that this may be because of reduced information asymmetries associated with U.S. investor protection regulations, including its accounting standards, disclosure rules, and security laws. This interpretation is consistent with the bonding hypothesis advocated by Karolyi (2012).

history and culture. For example, many emerging market firms are SOEs or family-owned businesses. Changing the management of these firms can be extremely difficult. Even if investors may successfully retire the incumbent management, it is very likely to result in the government intervention or management vacuum. Because rooted in the culture, history, and the environment of emerging market firms, the capital market segmentation is unlikely to disappear in a short-term period even after the cross-border listings.¹¹

The second challenge lies in the perspectives of information disclosure. To protect investors, developed stock markets impose stringent disclosure requirements and many emerging markets do not have the comparable reporting standard. For example, in the developed stock market, listed companies must regularly and comprehensively disclose all value-relevant information. However, it is not unusual for emerging market firms to withhold material information about earnings, debts, and investments owing to the personal interest of the top management (Goto, Watanabe, & Xu, 2008). In these cases, listing overseas means undergoing significant changes in disclosure environments. For emerging market firms, bonding with institutions in a different context may not be viewed as an advantage but as a misgiving, especially for firms in the countries where informational disclosure requirements are weak. In emerging markets, it is fairly common that the largest shareholder takes advantage of pyramidal or cross-holding structures to expropriate minor shareholders (Claessens et al., 2000; Lang, 2005). For these companies, listing overseas to a developed market means that they need to reveal their ownership structure to the public. If they do so, their chances to consume private benefits could be significantly reduced. Therefore, the gap between information disclosure policies imposed by the developed market exchanges and the amount of information that emerging market firms are willing to disclose can be so large as to impede their cross-border listing intentions.

The third challenge comes from country-level factors. The literature documents that political force significantly influences the decision making of emerging market firms. There are two relevant

¹¹ The ability of cultural and institutional adaptation is an important reason why newly established companies (such as technology firms) are relatively easy to list on the U.S. stock exchanges. In contrast, traditional emerging market firms (such as banks) are more difficult to conduct U.S listings.

arguments. The one argument states that overseas listing serves a policy tool. Under this argument, firms with political connections are more likely to list overseas because by doing so, the listed firms can attract political media and the managers of the listed firms can gain political rents for their future promotion (Hung et al., 2012; Sun et al., 2013). The other argument states that issuing shares to the public is a way of the privatization that facilitates the reform of SOEs (Boubakri, Cosset, & Saffar, 2013; Sun & Tong, 2003). This reform is important to SOEs because managers of privatized firms would be more careful about risk-taking activities and care more about earning quality. Both arguments suggest that the government is one of the major powers in emerging market firms regarding cross-border listings. Further, the cultural differences between the emerging and developed markets are usually large, which increases the degree of unfamiliarity and therefore significantly influences cross-border listing decisions (Pagano, Röell, & Zechner, 2002; Sarkissian & Schill, 2004). All these factors can form constraints that impede cross-border listings of emerging market firms.

2.2 The role of Institutional investors

Since the 1930s, institutional investors have gradually risen to a major countervailing power to corporate insiders (Farrar & Girton, 1981).¹² In view of this tendency, Farrar and Girton (1981) raise an important question: Whether the increased power of institutional investors benefit only themselves or all stakeholders in the market at large? Undoubtly, it is likely that the institutional investor may promote cross-border listing because the institutional investor constrains the private benefits taken by controlling shareholder, who will be equipped a stronger incentive to take bonding premium from cross-border listing. Nonetheless, recent studies provide general support that institutional investors help improve corporate governance, promote corporate information disclosure and bridge country-level differences, which will benefit all stakeholders in the market at large.

First, the function that institutional investors contribute to the improvement of corporate governance is

¹² An in-depth study on the growth of the U.S. financial market by Farrar and Girton (1981) shows that institutional investors had significantly increased corporate shareholdings since the 1930s. Managers responded to this situation in two ways. (1) Collaborating with institutional investors to avoid the possibility of confrontation. (2) Seeking to attract more individual investors to diffuse the shareholdings.

in line with the interests of all stakeholders as a whole. For example, Hartzell and Starks (2003) report that a greater institutional ownership concentration has a positive effect on the sensitivity of executive compensation to firm performance and a negative effect on the level of compensation. Aggarwal et al. (2011) and Kang et al. (2018) examine the relation between institutional holdings and forced CEO turnovers, finding that the higher the number of institutional blockholders, the more the likelihood of replacing poorly performed CEOs, the higher the stock returns around the forced CEO turnover announcements, and the better the performance after the replacement of poorly performed CEOs. However, the role of institutional investors in corporate governance is usually ignored. The reason for this is that they usually do so in a manner of behind-the-scenes (McCahery et al., 2016). In other words, institutional investors do not command firm operations directly. Instead, they focus more on firms' long-run strategies and use their voting rights to make an influence. As evidence, Aggarwal et al. (2015) show that institutional investors highly participate in the proxy process and tend to use voting rights to influence corporate governance. With this characteristic, institutional investors help overcome the myopic actions of managers and promote the investments that benefit for firm long-term development (Bena et al., 2017). Based on these studies, it is expected that institutional ownership facilitates cross-border listings of emerging market firms by aligning the governance standard differences between emerging market firms and developed markets.

Second, institutional ownership has also shown to promote information disclosure. Bird and Karolyi (2016) find that there is a strong positive effect of institutional holdings on both the quality and quantity of the disclosure. While it is likely that institutional investors have a connection with insiders, they trade on the different sets of information from that of the insiders (Griffin et al., 2012; Sias and Whidbee, 2010). This is the reason why institutional investors have a strong demand for corporate public information. Further, institutional investors not only have high demand information disclosure, but they also stress on the content and quality of information disclosure. Choi and Sias (2012) report that institutional investors respond strongly to analyst forecasts on corporate financial information. Jiang (2010) finds that institutional investors tend to incorporate intangible information¹³ into the stock price.

¹³ The information not associated with tangible factors such as firm size.

Further, using a stock market as a medium, Boehmer and Kelley (2009) find that the informational efficiency of stock prices is directly associated with institutional holdings, as firms with a higher degree of institutional holdings, their stocks are priced more informationally efficient. These studies confirm the information role of institutional investors.

Finally, institutional investors also play a role in bridging the country-level differences. Ferreira et al. (2009) examine the role of institutional investors in cross-border mergers and acquisitions (M&As) and find that foreign institutional ownership is positively related to the possibility, intensity, and success rate. This provides direct evidence that institutional ownership bridges country-level differences and therefore facilitates cross-border financial activities, and therefore bridges firms in different markets and promote the integration of world financial markets. Further, Dahlquist and Robertsson (2001) find that foreign institutional investors are more likely to have ownership of firms traded in the markets with great liquidity. These studies suggest that institutional investors play an important role in connecting cross-border financial activities when country-level differences are large.

Taken together, while cross-border listings are beneficial activities to emerging market firms, their managers and controlling shareholders may be reluctant to list overseas due to increased challenges. In contrast, institutional investors have a stronger tendency to carry out the equity listing and cross-border financial activities. Therefore, institutional ownership may serve as an important driving force for cross-border listings of emerging market firms.

3 Data

3.1 Sample selection and cross-border listing variables

To obtain the cross-border listing data of emerging market firms, a sample of Chinese firms carried out IPOs over the period from 1990 to 2017 is compiled. Data for cross-border listed firms are procured from the HKSE, which classifies different types of cross-border listings in a detailed way. On the HKSE's website, the "China dimension" section contains comprehensive information regarding Chinese firms listed on the HKSE, including that the share types of listed Chinese firms are clearly

defined as “H-shares” (shares are issued by firms that are incorporated in China mainland) or “Red-chip” (shares issued by firms incorporated overseas, though their main assets or revenue come from China mainland). This information is only available in the HKSE. Further, the HKSE’s website also clearly shows these listing channels. A Chinese firm can list in the HKSE via the mainboard and the Growth Enterprise Market (GEM). A firm needs to confirm a higher standard to list on the HKSE mainboard than to list on the GEM. Firms listed in the Chinese mainland are used as the reference firms and obtain data from the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). The information about listing dates and industry sectors is obtained through HKSE, SSE, SZSE, and Bloomberg.¹⁴

[Insert [Table 1](#) Here]

[Table 1](#) reports the distribution of the sample firms. Over the sample period, 259 firms are listed in Hong Kong. Among them, 146 firms (56.37%) are listed as H-shares and 113 firms (43.63%) are listed as Red-chips. In terms of listing channels, 231 firms (89.19%) are listed on the mainboard, and 28 firms (10.81%) are listed on the GEM. Panel A of [Table 1](#) reports the sample distribution by year, showing that in the early year, Chinese firms list on the HKSE predominantly using the way of Red-chips; but recently, they usually take the forms of H-shares.¹⁵ Further, throughout the sample period, the majority of Chinese firms choose the HKSE mainboard as the cross-border listing venue.¹⁶

Panel B of [Table 1](#) reports a sample distribution classified by industry sector. It shows that consumer (cycle and non-cycle), industrial and financial sectors cover the most listings for both H-shares (with 24.65%, 24.65%, and 23.29% of Hong Kong listings, respectively) and Red-chips (with 30.09%, 17.70%, and 23.01% of Hong Kong listings, respectively).¹⁷ Firms in these sectors also mainly choose

¹⁴ See https://www.hkex.com.hk/Market-Data/Statistics/Consolidated-Reports/China-Dimension?sc_lang=en for HKSE listings, <http://www.sse.com.cn/market/stockdata/raise/ipo> for SSE listings, and <http://www.szse.cn/main/en/marketdata/stockinformation> for SZSE listings.

¹⁵ It is worth noting that, private share (P-chip) firms may use a similar way as Red-chips to be listed on the HKSE. However, these firms are also not officially included in the China-dimension by HKSE, and therefore out of the scope of this study.

¹⁶ The peak of the GEM listing is in the 2001-2005 period when 19 sample firms choose GEM as the listing venue. Nonetheless, during this period, there are still more firms (45 sample firms) choose the mainboard as the listing venue.

¹⁷ Principally, H-share firms must be incorporated in China, however, the data in this study show that there is

to list on the HKSE mainboard instead of the GEM. In addition, unsurprisingly, the most common type of firms listed on the GEM comes from the technology sector.

Panel C of Table 1 reports the sample distribution by region. The five regions are classified according to the National Bureau of Statistics of China.¹⁸ Beijing is the capital city of China, firms headquartered in Beijing are likely to be subject to political influence, which has documented in the literature as one of the main driving factors for cross-border listings of emerging market firms (Hung et al., 2012; Sun et al., 2013). The data suggest that more than a third (34.25%) of H-shares are Beijing incorporated firms. This provides consistent evidence of the political argument of cross-border listings. Guangdong is the province that shares a similar culture and language (Cantonese) and is geographically adjacent to Hong Kong. Pagano et al. (2002) and Sarkissian and Schill (2004) document that geographical and cultural proximity significantly affects overseas listing decisions. Out of H-share listings, 16 sample firms come from Guangdong province, this number is similar to that of Middle and Northeast (16 firms) and West (18 firms) regions. The East region is the most economically developed areas of China. The data show that 30.83% of H-shares are incorporated in the East region. These figures seem to suggest that Chinese listings are prone to political and economic influences.¹⁹

one firm incorporated overseas and its address is “28 Queen’s Road Central, Central Tower, 4th-5th Floor, Central, Hong Kong (SAR)”. Closer scrutiny, the firm is “Zhejiang Chang'an Renheng Technology Co Ltd”. As its name indicated, while the firm’s office is located in Hong Kong, it was incorporated in Zhejiang province of China. Similarly, all Red-chip firms theoretically should be incorporated overseas, but they may be headquartered in the Chinese mainland as data suggested, there are 5 sample firms in Beijing, 2 sample firms in Guangdong, 7 sample firms in eastern regions, and 2 sample firms in the rest part of China.

¹⁸ This study lists firms headquartered in Beijing and Guangdong separately, as the former is likely to subject to political influence and the latter is culturally close to Hong Kong. Besides these two regions, the East region includes Fujian, Hainan, Hebei, Jiangsu, Shandong, Shanghai, Tianjin, and Zhejiang; the Middle region includes Anhui, Henan, Hubei, Hunan, Jiangxi, and Shanxi; the Northeast region includes Heilongjiang, Jilin, and Liaoning; and the West region includes Chongqing, Gansu, Guangxi, Guizhou, Neimenggu (Inner Mongolia), Ningxia, Qinghai Shanxi, Sichuan, Xinjiang, Xizang (Tibet), and Yunnan. For the detail, see the website of the National Bureau of Statistics of China:

http://www.stats.gov.cn/ztc/zthd/sjtr/dejtkfr/tjqp/201106/t20110613_71947.htm.

¹⁹ Similar to the Hong Kong-listed sample, the vast majority of Chinese mainland-listed firms are from the consumer and industrial sectors, with 33.39% and 32.01% of the overall domestic listings, respectively. Unlike Hong Kong listings, however, the financial industry is not the main component of domestic listings, with only 5.83% of the overall sample domestic listings. This figure is a sharp contrast to that of cross-border listings. Interestingly, for domestically listings, Beijing is not the most common location where listed firms come from. Mere 15.14% of sample firms are headquartered in Beijing. In contrast, 41.23% of sample firms are headquartered in the East region, meaning that the eastern region hosts most of the domestically listed firms. In addition, 18.17% of sample firms are headquartered in Guangdong. These numbers suggest that regional characteristics can be influential factors for domestic listings.

One special kind of cross-border listings is the firms listed in both the Chinese mainland and Hong Kong, so-called “AH-shares”.²⁰ Table 1 shows that in the early years, Chinese firms commonly adopt the AH-listing approach when they go overseas. But more recently, they tend to directly list on the HKSE.²¹ Further, the majority of “AH-share” firms come from the consumer (24.49%), financial (25.51%), and industrial (28.57%) sectors. In addition, Beijing-based firms play a significant role in the AH-shares with 34.69% of the total AH-listings, followed by the East region with 27.55% of the total AH-listings. These proportions are close to those of cross-border listings as a whole.

3.2 *Variable construction*

3.2.1 *Institutional ownership*

To investigate whether and how institutional investors affect cross-border listings, this paper focuses on strategic ownership. To ensure that cross-border listing decisions have a material influence on the investments made by institutional shareholders, this paper rules out the speculative tradings, the trading only for short-term profits, and the trading that is negligible to listing decisions. Further, to make sure that institutional holdings are not restricted and thus the institutional shares are not forced to be held, data on the free float shares are employed, with which institutional investors can buy or sell freely. That is, if the cross-border listing decision made by a firm disagrees with the strategy of institutional investors who hold the shares of the firm, they can freely sell their holdings. Therefore, the institutional investors in this study not only are interest related to the firm but also influence cross-border listing decisions.²²

²⁰ For Chinese firms, cross-listing in both markets is only possible for H-shares (not possible for Red-chips as these firms are registered overseas) and predominantly traded on the mainboard of the HKSE. The SSE’s website provides information regarding “AH-shares”.

<http://www.sse.com.cn/assortment/stock/areatrade/ahassortment>. It is also available on the HKSE’s website (http://www.hkex.com.hk/chi/Invest/misc/documents/ic_for_ah_shares_tc.pdf). This therefore confirms the Chinese AH-listing data with the information on both stock exchanges.

²¹ It is noteworthy that in the 2006-2010 period, the number of cross-border listings (pure Hong Kong listings plus AH-listing) exceeds the number of pure domestic listings. One of the important reasons is that in this period, the China Securities Regulatory Commission (CSRC) froze domestic IPO and stopped the approval of any IPOs to prevent the stock market from further plunging. Due to the impact of the global financial crisis, the SSE Composite Index plunged from its historical high 6,124 points in 2007 to 1,664 points in 2008. In this environment, CSRC initiated the Share-trading Reform of Listed Companies and proclaim that a stable stock market is the pre-condition for the reform. This meant that the IPO was possible only when the reform was completed.

²² It is worth noting that strategic institutional shareholders may be nor may not the controlling shareholders. Further, prior to the IPO, the information about when the strategic institutional investors join the company is not

Based on these definitions, two variables are constructed to measure institutional ownership, each of which captures specific aspects of institutional ownership, to ensure that the empirical tests are not biased to a specific measure of institutional ownership. The first variable is the percentage of strategic free float shares held by institutional investors (*Institutional Holding*). The percentage of shareholdings is the most straightforward measure of ownership and is commonly applied in the literature (Claessens et al., 2000; Doidge et al., 2007; Lemmon & Lins, 2003). Further, as not all listed firms have institutional investors, the second way to capture the effect of institutional ownership is a dummy variable (*Institutional Index*) that equals one if at least one strategic institutional investor appears in the sample form, and otherwise zero. This measure emphasizes how the presence of institutional investors influences cross-border listings. The primary benefit of using these two institutional ownership proxies is that they estimate not only the effect of the size of institutional shareholders' stakes but also the importance of having strategic institutional investors. The data used for constructing these two measures of institutional ownership are compiled from the Datastream.

3.2.2 Control variables

Throughout this paper, the empirical analyses include a set of the firm-level and regional-level control variables that have been documented in the literature to have effects on cross-border listing decisions. First, as in Doidge et al. (2009), the following firm-level variables are included — Return on assets (*ROA*), *Sales growth*, *Firm size*, *Offer size*, *Leverage ratio*, and *Financial flexibility index*. *ROA* is a proxy for firm operating performance and profitability, measured by the ratio of net income to total assets. Firms with better performance are expected to be more likely to list overseas in order to externalize their earnings advantage. *Sales growth* captures operational growth and is calculated as the annual percentage change in sales revenue. The rationale for controlling operational growth is that if cross-border listings can provide resources to support growth, then the firms with higher operational growth are expected to have a strong tendency to list overseas. *Firm size*, computed by the natural logarithm of total assets, is a proxy for the effect of the economic scale. Large firms are more likely to

required to disclose. The bottom line is that at the time of IPO, we can clearly identify a firm's institutional ownerships and that at the time of cross-listing, the institutional shareholders must hold at least 5% free-float shares for long-term strategic reasons.

obtain resources to support their cross-border listings. Further, it is well documented that the pricing mechanism is different between the Hong Kong stock market and China mainland markets. *Offer price* is used to control this effect and is measured by the stock price initial offer price in Hong Kong dollars. The data for *Offer price* is procured from Bloomberg. *Corporate leverage* is measured as the ratio of total debt to total capital to control for the effect of capital structure. It may argue that firms with a higher level of debt are more likely to issue equity overseas to balance the capital structure. This paper also controls for *Financial flexibility*, as the literature indicates that one of the main motivations for a firm list overseas is to relax financial constraints (see, for example, Lins et al., 2005). *Financial flexibility* is measured using the approach introduced by Doidge et al. (2009).²³

Recent studies highlight the influence of geographical and regional characteristics of cross-border listings (Pagano et al., 2002; Sarkissian & Schill, 2004), the paper therefore employs a number of variables to control for geographic and regional effects. First, Doidge et al. (2009) argue that firms are more likely to carry out cross-border listings if the firm is more economically open. It is therefore needed to control for the degree of openness of sample firms. One argument is that firms are more open to foreign trade may be more likely to conduct cross-border financial activities as well. Alternatively, firms with high growth but less open to foreign trade may use the cross-border listing as a way to establish a connection with foreign markets. The proxy for *foreign openness* is the natural logarithm of the total foreign trade volume of the province where a sample firm is headquartered, as reported by the Ministry of Commerce, the People's Republic of China.²⁴ Further, similar to Doidge et al. (2009), this paper also controls for regional economic development measured by the natural logarithm of Gross Domestic Product (*GDP*) of each province, the data are provided by the National Bureau of Statistics of China.²⁵ Finally, as noted in Doidge et al. (2009), firms are more likely to list on a foreign market where social development is close to the local market. To capture this effect, the difference in the

²³ In this approach, the financial flexibility index is constructed according to capital expenditures, dividend ratios, and cash and equivalents. The maximum value of this financial flexibility index is three, if a firm is below the 25th percentile of capital expenditures, above the 75th percentile of dividend ratios, and above the 75th percentile of cash and equivalents, as each of these criteria gives one point to the firms. On the other hand, the minimum value of the financial flexibility index is zero.

²⁴ <http://english.mofcom.gov.cn/article/statistic>.

²⁵ <http://data.stats.gov.cn/english/easyquery.htm?cn=E0103>

Human Development Index (*HDI*) between Hong Kong and the firm headquarter location is calculated. The information for the *HDI* is from the China National Human Development Report and the United Nations.²⁶ The detailed description and data sources for these variables are summarized in Appendix 1.

3.3 Summary statistics

[Insert [Table 2](#) Here]

[Table 2](#) reports the summary statistics of the variables used in this study.²⁷ The first two rows of [Table 2](#) present the measures of institutional ownership. *Institutional Holding* has a mean value of 0.72% (the maximum value is 54.67%), suggesting that Chinese listed firms on average have 0.72% free float shares held by strategic institutional investors. For the firm with the highest institutional strategic holding, the percentage of this ownership is 54.76%.²⁸ The mean *Institutional Index* is 0.21, suggesting that on average 21% of Chinese listed firms have strategic institutional investors. This reflects that having strategic institutional investors is not very common for the Chinese listed firms.

The next ten rows (from the third row to the twelfth row) of [Table 2](#) report summary statistics for control variables. In terms of firm-level controls, the average *ROA* of sample firms is 9.69%, which is comparable to 9% reported by Sun and Tong (2003). The average sales growth is 20.18%, comparable to the numbers in Busaba et al. (2015) who document a mean sale growth of 21.23% for dual-listings and 23.38% for domestic listings. The average firm size is 21.30 (the natural logarithm of the book value of total assets), which is comparable to the number reported in Li et al. (2015). As for other firm-level controls, the average leverage is 16.72%, the average offer price is \$11.26, the average offer size is \$650 million, and the average value for *Financial Flexibility Index* is 1.24. In terms of country-level

²⁶ http://www.cn.undp.org/content/china/en/home/library/human_development.html

²⁷ These variables can be broken out into four groups: The measure of institutional ownership, the key independent variables of interest, control variables, and variables used for the further tests (these variables include *Analyst coverage*, *Earning surprise*, *Forecast dispersion*, *Market to book value*, and *Government ownership*, which will be discussed in Section 4.3.1).

²⁸ When comparing these numbers with *Government Ownership*, which are reported in the last row of Table 2, it can be seen that the government ownership is much higher (the mean value of *Government Ownership* is 43.44%) The comparison reveals the depth of government holdings in Chinese listed firms. The effect of government ownership will be discussed in detail in Section 4.2.

controls, the sample average of *Log of foreign trade* is 22.84, slightly lower than China's overall average of 27.52. The mean *Log GDP* for the sample firms is 11.30, slightly higher than China's overall average of 10.996. The sample average *Difference in HDI* is 0.12, which is smaller than the difference between China's average level and Hong Kong's HDI.²⁹ Overall, the summary statistics of the variables in this study is generally consistent with previous Chinese and cross-border listing literature.

Pairwise correlation coefficients (untabulated for the sake of brevity) indicate a moderate correlation of 0.53 between two measures of institutional ownership (*Institutional Holding* and *Institutional Index*), suggesting they capture non-identical aspects of institutional ownership. The correlations between institutional ownership measures and other variables are also in general low. For example, the highest correlation for *Institutional Holding* is with *Firm Size* at 0.22 and the highest correlation for *Institutional Index* is also with *Firm Size* at 0.20. Therefore, multicollinearity is not a particularly serious issue here. This allows carrying out regression analyses to empirically examine the influence of the institutional ownership on the cross-border listings. The results are reported in the subsequent section.

4 Results

4.1 *Institutional ownership and the cross-border listing decision*

This section carries out empirical examinations on the relation between institutional ownership and the probability of cross-border listings. Specifically, it investigates different types of listings, addresses endogeneity issues, and implements exogenous shocks to ensure the robustness of the results.

4.1.1 Multi-variant logistics regression results

Emerging market firms are subject to lower quality of corporate standards, and cross-border listings offer them an important channel to improve their standards. To distinguish the effect of institutional ownership in terms of different listing types, the sample is classified based on the listing channels, and it contains three subgroups: H-share listings, Red-chip listings, or AH-listings. Alternatively, the tests

²⁹ In 2017, the HDI of Hong Kong is 0.933 and the average HDI of the Chinese mainland is 0.752, and the difference is 0.181.

are also conducted based on the venues of listings, and it contains two subsamples: Mainboard listings and GEM board listings. Mainboard listings enforce stricter disclosure and governance requirements than GEM board. The effect of institutional holdings is expected to be more significant in listings as H-shares and on the mainboard, as these types of listings impose even higher requirements by HKSE.³⁰ The standard multinomial logistic regressions with the different groups of cross-border listings are estimated. We control for the firm- and country-level variables and industry fixed effects throughout all the analyses because cross-border listings demonstrate strong industry pattern.³¹ The reference for all types of cross-border listings is domestic listings. The standard errors are clustered at the firm level.³²

[Insert [Table 3](#) Here]

Panel A of [Table 3](#) presents results using *Institutional Holding* as the measure of ownership of institutional investors. It shows that the coefficients on *Institutional Holding* are positively related to all five types of cross-border listings. These results suggest that the percentage of the institutional holding relates to the probability of all types of cross-border listings relative to domestic listings. Among them, four out of five coefficients are significant at the 1% level, except only for the Red-chip regression. Given that the approach used in the examination is multinomial logistic regressions, the coefficients obtained only can be interpreted in terms of signs and the significance level, but not for their size. To understand the economic significance, the marginal effect therefore is calculated. In H-shares regression, the marginal effect for *Institutional Holdings* is 0.009 with a t-statistic of 4.17, suggesting that a 1% increase in institutional holding corresponds to a nearly 1% higher probability of listing as H-shares, relative to that of listing domestically. This effect is significant at the 1% level. Likewise, the marginal effects for *Institutional Holdings* in the regressions of AH-listings, Mainboard and GEM listings are all positive and statistically significant. These results suggest that a 1% increase

³⁰ For firms to be listed as H-shares, they need to confirm both corporate laws of the Chinese mainland and listing regulations of HKSE.

³¹ It is noteworthy to point out that government and political influences may play a key role in the oversea listing which requires the endorsement by the central government, who may be more likely to favor SOEs. However, the available data regarding government ultimate ownership is limited. Controlling government ownership would significantly reduce the degree of freedom of our analyses in this section. Nonetheless, we do focus on tackling the issue regarding the identity of state-owned enterprises in Section 4.2.

³² The result is highly consistent when clustering standard errors at the industry-year and regional level.

in *Institutional Holding* corresponds to 0.30%, 0.80% and 0.14% of a higher probability of the AH-listing, the listing on the mainboard, and the listing on GEM, respectively, relative to that of listing domestically.³³ Overall, the results also indicate that a higher percentage of *Institutional Holding* is associated with stronger effects in promoting cross-border listings, in particular H-share listings and mainboard listings, two main streams of listings on the HKSE with stricter listing requirements.

Panel B of [Table 3](#) presents results in which *Institutional Index* is used as a proxy for institutional investors' strategic ownership position. It is shown that the coefficients on *Institutional Index* are positive and significant in three out of five regressions, except for the Red-chip and GEM regressions. The result suggests that having a strategic investor(s) increases the probability for a firm to carry out the main types of cross-border listings. Especially, in the H-shares regression, the marginal effect of *Institutional Index* is 0.0545, suggesting that having strategic institutional shareholders corresponds to a 5.45% higher probability of listing as H-shares, relative to that of listing domestically. This effect is significant at the 1% level (t-statistic of 5.16). Similarly, the presence of strategic institutional ownership corresponds to 3.59% and 6.02% higher probability of listings as AH-listings and listings on the HKSE mainboard, respectively, relative to that of purely domestic listings. Again, the results demonstrate that firms with strategic institutional ownership are more likely to list as H-shares and on the mainboard of HKSE.

Control variables in general show predicted signs as well. For example, [Table 3](#) shows that the firm-level characteristics appear to be more important in explaining cross-border listing decisions. Particularly, the coefficients on *ROA* and *Financial flexibility* are significantly positive related to Red-chip listings, suggesting that firms with better operational performance and less financial pressure are more likely to carry out listings for oversea incorporated firms. This finding is consistent with Doidge et al. (2009). Specific to Chinese cross-border listings, *Firm size* is positively and significantly associated with the H-share listings, the Red-chip listings, and AH-listings, but negatively and significantly associated with the GEM listings. This should not be a surprise as large firms tend to list

³³ In the Red-chip regression, *Institutional Holding* does not demonstrate significant results.

as H-shares, Red-chips, and AH-shares, and small firms tend to list on the GEM board. Further, there are consistently negative coefficients on *Offer price*. This is in line with empirical findings that Chinese cross-border listings are associated with a valuation discount (Arquette, Brown, & Burdekin, 2008; Chan, Menkveld, & Yang, 2008).

In terms of regional-level factors, Panel A of [Table 3](#) shows that *Log foreign trade* is negatively and *Log GDP* is positively related to the GEM listings, implying that firms located in the regions with lower levels of openness and higher degrees of economic development are more likely to list on the GEM board. However, there is no significant evidence that regional variables directly affect cross-border listing decisions on the main types of listings (H-shares listings and Mainboard listings).

Taken together, the results in this subsection provide supporting evidence that institutional holdings are positively associated with cross-border listings of Chinese firms, suggesting that institutional investors play an important role in overseas listing decisions of emerging market firms. This effect is particularly stronger for the main types of listings (H-share listings and the mainboard listings).³⁴

4.1.2 The two-stage least squares (2SLS) instrumental variables approach

The following subsections address endogeneity issues. The existing literature shows that institutional investors prefer stocks of the companies (and in the countries) with high corporate governance standards and listed on the exchanges with stringent disclosure requirements. Considering that the HKSE requires more stringent disclosure requirements as well as higher corporate governance standards than mainland Chinese stock exchanges in Shanghai and Shenzhen, it seems possible that institutional investors are simply attracted to those Chinese stocks that are (or expected to be) cross-listed on the HKSE. In other words, the implied causality may run in the opposite direction. It is therefore important to formally establish the causality after controlling for the endogeneity of institutional ownership.

³⁴ It is worth noting that in Panel A of Table 3, institutional holding is not significantly related to Red-chip listings. In panel B, it turns positive and significant. This suggests that the presence of institutional investors has significantly positive effects, regardless of the exact proportion of holdings. In contrast, the presence of institutional investors does not make a firm more likely to list on the GEM, but the portion of holding matters more for a GME listing decision. For the other three main types of listings (H-share listings, AH-listings, and the mainboard listings), there is a significant effect for both the proportion and the presence of institutional investors.

To address reverse causality issues, we follow the literature, such as in Chung, Elder, & Kim (2010), to employ the 2SLS method. The 2SLS regressions use instrumental variables that are related to institutional ownership, but unlikely to be correlated with residuals in the second-stage regressions.³⁵ The literature suggests that the fundamental motivation for institutional investors to participate in the IPO activities is that they can reap considerable profits, which is related to the size of stock issuance (Chiang, Qian, & Sherman, 2009; Jenkinson & Jones, 2008). Chemmanur et al. (2009, 2010) report that institutional investors are better informed and can obtain benefits from such information. In other words, the larger the stock issuance, the higher the profitability, the more the likelihood that institutional investors participate in the stock issuance. Based on this justification, it is reasonable to use cash acquired through IPO as an instrumental variable for institutional ownership. We capture issue size by the natural logarithm of the Hong Kong dollar amount of *Offer size*.

[Insert [Table 4](#) Here]

[Table 4](#) presents the results of the probit model with instrumental variables regressions estimated by the 2SLS regressions. The first stage regresses two variables discussed previously for institutional ownership on *Offer size* and all of the other variables in the second stage regression. The second stage regresses the likelihood of cross-border listings on the predicted values of institutional ownership (from the first-stage regression) and all other explanatory variables. The dependent variable is a dummy variable that equals one for cross-border listings and zero otherwise (i.e. domestic listings). Columns (1) and (3) of [Table 4](#) present the results of the first-stage regressions, and columns (2) and (4) give the results of the second-stage regressions. Both the first- and second-stage regression results are based on clustered standard errors at the firm level. Firm dummy and year dummy are included in these regressions to control for the impact of macroeconomic fixed effects.

To assess the strength of the instrumental variable, standard tests for weak instruments suggested by Stock, Wright, & Yogo (2002) are performed. First, the *F-statistic* of the first-stage regressions with

³⁵ The fixed effect model is not feasible in this case due to a small proportion of institutional ownership that does not significantly change over time.

the instrument is computed. The result of *F-statistic* demonstrates a joint statistical significance of the instrument, as it rejects the null hypothesis that the coefficients on the instruments are jointly zero. Further, the partial R^2 s add about 31.63% (in Column 1) and 15.02% (in Column 3) to the R^2 in the first-stage regressions, respectively. These tests suggest that the instrument employed is not weak. It is also worth noting that the *Wald test* of exogeneity gives significant χ^2 in both regressions, which reject the null hypothesis of exogeneous regressions and suggest that endogeneity is indeed an issue in this context and that the instrumental variable used here is a valid approach.

Consistent with the conjecture, the results in Columns (1) and (3) show that the estimated coefficients on the instrument variable (*Offer size*) have the expected positive sign and are highly significant. These results confirm that institutional investors tend to have ownership of large stock offers. More important, in the second-stage regressions, the results show that the coefficients on the instrumented *Institutional Holding* and *Institutional Index* are statistically significant and positive, indicating that the positive effect of institutional ownership on cross-border listings remains intact even after controlling for the potential reverse causality problem. Both coefficients on *Institutional Holding* (in Column 2) and *Institutional Index* (in Column 4) are statistically significant. It is noteworthy that the estimated coefficients increase in magnitude relative to the results in the previous subsection. This can be attributed to the instrumental variables approach reducing the errors-in-variables bias and reconfirms the strong positive relation between institutional ownership and cross-border listings.

The results for other variables are also worth noting. After addressing the reverse causality, firm leverage remains significantly negative, indicating that firms with a higher debt ratio are less likely to list overseas relative to list domestically. The result stresses the importance of financial health in cross-border listings. Nonetheless, other firm-level variables become insignificant. In contrast, regional factors turn to significant. Specifically, *Difference in HDI* and *Log GDP* are significantly positive, suggesting that firms in the regions where better economically developed and less socially developed have a stronger tendency to list in the HKSE. Considering the literature also suggests that Chinese government policies have a strong influence on firm cross-border listing decisions, the results indicate that the external environment has a considerable impact on Chinese firm listing decisions.

Taken together, the results in this subsection using 2SLS regressions do not alter the positive relation between institutional ownership and the possibility of cross-border listings. This suggests that the relation found previously can be interpreted causally, that is, the greater the institutional ownership of a Chinese firm, the higher the likelihood of cross-border listing of the firm.

4.1.3 Results with propensity score-match

To examine the effect of institutional ownership on cross-border listings, a test with a propensity score-matched sample is employed. As the mean institutional ownership in its sample firms is only 0.72% and almost 80% of sample firms do not have institutional investors at all,³⁶ this approach would help address the concern that observable firm-specific characteristics affect the probability of firms being listed on the foreign exchanges. Specifically, the firms with institutional investors are matched with firms without institutional investors by the firm and regional characteristics that have been discussed above. After matching, the differences between domestically and foreign listed firms are investigated. To ensure that the result is not driven by a certain propensity score-matching methodology, four different techniques are employed, including *the Nearest-neighbor matching*, *the Radius matching*, *the Kernel matching*, and *the Stratification matching*.

[Insert [Table 5](#) Here]

[Table 5](#) reports testing results from propensity score-matching techniques. Panel A of Table 5 shows the summary statistics of 524 treated observations (firms with institutional investors) and 2,020 controlled observations (firms without institutional investors). This means that the sample is appropriate to carry out propensity score-matching tests, as it provides sufficient firms to find good matches for treated observations. Panel B of [Table 5](#) presents the results from the first-stage of propensity score matching tests, i.e. the probit model. The dependent variable is the dummy variable of treated observations, that is, whether or not the firm has institutional investors. This stage generates propensity

³⁶ It is worth noting that to be included in the sample, it is required that an institutional investor must hold strategic free float shares. Put it another way, even if a firm simply has institutional shareholders but they are not strategic or the shares that they hold are not free float, the firm is not included in the sample.

scores that will be used for matching in the second stage.

Panel C of [Table 5](#) reports the average treatment effect of treated (ATET) with four propensity score-matching techniques: *ATET Nearest-neighbor*, *ATET Radius*, *ATET Kernel*, *ATET Stratification*. The results show that the coefficients are significantly positive throughout four propensity score-matching techniques. After matching treated and control firms, the effects of institutional ownership are to increase the probability of cross-border listings by about 19.9% (*ATET Nearest-neighbor matching*) to 30.7% (*ATET Radius matching*). In other words, firms with institutional ownership have a higher probability of cross-border listings than those that do not have institutional ownership. Therefore, even after addressing the effect of omitted variables, institutional ownership still shows a strong association with cross-border listings.

4.1.4 Results after implementing exogenous shocks

This subsection uses the passage of regulations associated with the structure change of the Chinese market as exogenous variations to test the effect of institutional ownership. There are three regulations closely related to institutional ownership and cross-border listings of Chinese firms. The first government policy that formally and comprehensively regulates Chinese cross-border listings was issued in 1997.³⁷ Then, in 2004, the State Council issued the “*Several Opinions on Further Promoting the Healthy Development of the Capital Market*”, declaring the enhancement and strengthening of institutional investors. In the same year, the China Securities Regulatory Commission issued a policy to standardize cross-border listing of subordinated enterprises of domestically listed companies³⁸. The latest document regulating cross-border listings was issued in 2012,³⁹ which provided regulatory

³⁷ The title of the document is “Circular of the State Council Concerning Further Strengthening the Administration of Share Issuance and Listing Overseas” and it can be found through the website: http://www.csrc.gov.cn/pub/csrc_en/OpeningUp/RelatedPolicies/OverseasListing/201211/t20121108_216652.html.

³⁸ The title of the document is “Circular of the China Securities Regulatory Commission on Several Issues Concerning the Standardization on Overseas Listing of Subordinated Enterprises of Domestically-Listed Companies” and the document can be found through the website: http://www.csrc.gov.cn/pub/csrc_en/OpeningUp/RelatedPolicies/OverseasListing/201211/t20121108_216654.html.

³⁹ The title of the document is “Special Provisions of the State Council Concerning the Floatation and Listing Abroad of Stocks by Limited Stock Companies.” and can be found through the website: http://www.csrc.gov.cn/pub/csrc_en/OpeningUp/RelatedPolicies/OverseasListing/201210/t20121024_216120.html.

guidelines on overseas stocks and listing documents and review procedures by the government. As these regulations are exogenous to institutional investors, if institutional ownership plays a significant role in cross-border listings, their effect is expected to be reinforced after the passage of these regulations.

To examine whether the effect of institutional ownership on the cross-border listing is sustained undergoing these regulations, a probit model with a difference-in-differences (DID) test is performed. Specifically, the following regression is estimated,

$$CBListing_i = \beta_0 + \delta_0 post97/post04/post12 + \beta_1 IOwnership_i + \delta_1 post97/post04/post12 \cdot IOwnership_i + Controls_i + \varepsilon_i \quad (1)$$

where *CBListing* is a dummy variable that sets one if cross-border listings and otherwise zero, *post97/post04/post12* are dummy variables used to capture the effect of regulation passage and equal to one for observations after 1997, 2004, and 2012, respectively, *IOwnership* is institutional ownership and is captured by *Institutional Holding* and *Institutional Index* as above, and *Control* includes all control variables as used in previous sections. The intercept, β_0 , is the average likelihood of cross-border listing of Chinese firms without institutional ownership before the passage of regulations. The parameter δ_0 captures the status of all Chinese listed firms after the passage of regulation. The coefficient on *IOwnership*, β_1 , measures the institutional ownership effect that is not due to the passage of regulations. The parameter of interest is on the interaction *post97/post04/post12 · IOwnership*: δ_1 measures the cross-border listing due to the institutional ownership as a result of the passage of regulations, controlling for other reasons and industry-level shocks. The standard errors in all regressions are clustered at the firm-level.

[Insert [Table 6](#) Here]

The estimate of Equation (1) is given in [Table 6](#).⁴⁰ Column (1) includes *Institutional Holding* and the passage of the 1997 regulation. The coefficient on the interaction is significantly positive, suggesting

⁴⁰ For brevity, the results for constant terms are not reported. However, unsurprisingly, these results are all significantly negative, suggesting that on average, Chinese firms without institutional ownership are less likely to carry out cross-border listings before the passage of regulations.

that the firms with a great share of institutional holdings are highly likely to carry out cross-border listings after the passage of the 1997 regulation. Column (2) and (3) give the same results, indicating that institutional holdings are also strongly associated with the passage of the 2004 and 2012 regulations, respectively, to conduct cross-border listings. Columns (4) to (6) use *Institutional Index* to capture the effect of institutional ownership and offer the same results with those in Column (1) to (3), except only for that the coefficient on the interaction term is not significant for the passage of 1997 regulation (Column 4). Nonetheless, the passage of 2004 and 2012 regulations are strongly related to the presence of institutional investors in promoting cross-border listings. These results suggest that the passage of these regulations indeed strengthens the role of institutional ownership in cross-border listings.

Further, there is no evidence that the passages of these regulations have positive effects on overall cross-border listings. The coefficients on the dummy variable of *post97/post04/post12* are insignificant in five out of six regressions and even negative in Column (5), which is associated with the passage of the 2004 regulation. Therefore, while the passage of these regulations is positively related to the effect of institutional investors, it cannot be generalized that it also strengthens the effect of all other kinds of investors for promoting cross-border listings. In addition, the coefficients on institutional ownership are significant in five out of six regression and control variables show consistent signs with the previous results, suggesting that institutional ownership facilitates cross-border listings before the regulation changes. Overall, this section suggests that the effect of institutional ownership withstand the exogenous shocks, which heighten instead of lowering the effect of institutional ownership on cross-border listings.

4.2 The government and political influences

This section focuses on an important issue to the Chinese market — if government considerations and political policies can influence the effect of institutional ownership on cross-border listings. It is well documented in the literature that government policies drive Chinese cross-border listings (see, for example, Hung et al., 2012). In such an environment, one may argue that institutional ownership appears to be too small to start a serious drive toward overseas listing. In this section, three approaches are adopted to address this issue.

4.2.1 Evidence from the firm headquarter locations

This subsection carries out the empirical examination by region. Firms incorporated in Beijing are usually saturated by government policies and thus their stock listing decisions may be subject to political influence. Further, Guangdong is culturally and geographically close to Hong Kong. These similarities may also have effects on cross-border listing decisions (Hung et al., 2012; Sarkissian & Schill, 2004). In addition, while China is a developing country, its economic developments are geographically unbalanced: Its eastern coastal region is better developed than the rest of China. If political and other characteristics specific to a region drive the main result, then it would be no consistently significant relation between institutional ownership and cross-border listings once estimate regressions by regions.

[Insert [Table 7](#) Here]

[Table 7](#) reports the descriptive statistics of institutional ownership across types and headquarter regions. Indeed, when splitting the sample based on listing venues, Beijing-headquartered firms with higher institutional holdings are listed through H-shares and on the mainboard of the HKSE. Whereas for the GEM, the highest institutional ownership appears in Guangdong headquartered firms.⁴¹ On average, institutional holdings are 0.36% of domestic listed firms, 2.28% of Hong Kong-listed firms and 2.10% of AH-listed firms. The issue reflected in [Table 7](#) regarding government and political influences may be that the government decides which companies should be cross-listed first and then over time, the cross-listed companies may attract more institutional investment as institutions prefer overseas-listed stocks. The importance of political connection as a determinant of Hong Kong listings can be inferred from the proportion (4.14%) of Beijing-based firms that are listed on the HKSE, which is much higher than the proportion (2.28%) of firms from all regions of China, as shown in [Table 7](#). Obviously, Beijing-based firms should be more deeply and more often politically connected than firms from other regions.

[Table 7](#) also shows that in both domestically and Hong Kong-listed firms, institutional investors are the most likely to appear in firms headquartered in Beijing and are the least likely in firms in the western

⁴¹ As “AH-share” firms, institutional ownership is the highest for firms headquartered in the eastern region. Throughout all the regions, institutional holdings are at least four times higher in the cross-border listings (pure Hong Kong-listings and AH-listings) than in the domestic listings.

region. Overall, [Table 7](#) shows that Hong Kong-listed firms and AH-listed firms are associated with higher strategic institutional ownership than domestically listed firms. To investigate the idea that regional factors drive the positive effect of institutional ownership on cross-border listings, a subsample analysis by regions is carried out. If firms' cross-border listing decisions are related to regional specific characteristics instead of institutional ownership, the positive effect of institutional holdings on cross-border listings would disappear once separately looking at each region.

[Insert [Table 8](#) Here]

[Table 8](#) estimates probit regressions in which the dependent variable is a dummy variable that equals one if a firm is cross-border listed and zero otherwise.⁴² Panel A of [Table 8](#) reports regression results for the sample firms headquartered in Beijing. Columns (1) and (2) report probit regressions with *Institutional Holding* and *Institutional Index* for all cross-border listings. The coefficient estimates on both *Institutional Holding* and *Institutional Index* are significantly positive, which is consistent with the previous results. While the coefficient on *Institutional Holding* is insignificant, providing supporting evidence that government policies influence the effect of institutional ownership, the coefficient on *Institutional Index* is significant at the 1% level. To explain the economic meaning of the results, the odds ratios from logistic regression are also calculated. The odds ratios for *Institutional Index* in Column (2) are 24.40, suggesting that for firms from Beijing and having institutional strategic ownership, the odds of listing overseas are twenty-four times as large as the odds for those firms without institutional strategic ownership. Therefore, even given government influence, institutional ownership still plays a significant role in cross-border listings.

Panels B, C, D, E of [Table 8](#) reports regression results for the sample firms headquartered in Guangdong,

⁴² Since it is already known from the above subsections that institutional ownership is more strongly related to the probability of the main types of cross-border listing, i.e. the H-shares listing and the mainboard listing, compared to other types of cross-border listings (Red-chip listings, AH-listings, and GEM listings), the focus of the analysis in this subsection is therefore the cross-border listings as a whole. To implement this analysis, the result for each region is largely the same as the results reported here, where H-share listings and mainboard listings demonstrate stronger effects. The result is significant after the inclusion of Red-chip listings, AH-listings, and GEM listings. This strengthens the argument that institutional ownership is positively associated with the probability of having a cross-border listing.

and eastern, middle and northeastern, and western regions of China, respectively. In all regressions, the coefficients on *Institutional Holding* and *Institutional Index* are positive and are significant in six out of eight regressions. These results suggest that for firms in these regions, with less political influences, institutional ownership is more important for cross-border listings. Particularly, in the province of Guangdong, where culture is similar to Hong Kong, institutional ownership still plays a significant role. Throughout all regions except for western region, *Institutional Index* demonstrates a stronger effect compared to *Institutional Holding*, suggesting that the presence of strategic institutional investors is more crucial to cross-border listing decisions compared to the exact proportion of their shareholdings.

In sum, this subsection infers that institutional ownership is strongly and positively associated with the probability of cross-border listings of Chinese firms across regions.⁴³ Therefore, while there is evidence of political influences on cross-border listings, the argument that government policies corrode the effect of institutional ownership on cross-border listings cannot obtain the overall support.

4.2.2 Evidence from foreign institutional shareholders

Another concern regarding political influences on the main result is that at least some of the domestic institutional investors, such as banks and insurance companies, may not be politically independent. If this is the case, the positive effect of institutional shareholders documented above is driven by domestic institutional investors, which eventually represent political influences. Further, comparing to foreign institutional investors, domestic institutional investors arguably understand better about Hong Kong's culture and language, so they are expected to be easier to adjust to the information transformation. This may drive firms with institutional ownership to choose Hong Kong as the listing destination. To address these concerns, this section carries out empirical tests by focusing on foreign institutional ownership. Prior studies indicate that foreign institutional investors exert a

⁴³ The results do reveal that to some extent, there are differences in the effect of institutional ownership in different regions. For example, cross-listings from the eastern region, where is more economically advanced, show more significant results in both *Institutional Holding* and *Institutional Index*. Further, subsample data by region also reveal that the existence of institutional investor(s) is critical for Beijing and Guangdong firms; whereas the holding proportion is more important for firms in other regions, especially for the western region firms.

strong effect on cross-border listing decisions (Bena et al., 2017; Ferreira & Matos, 2008).⁴⁴ It is unlikely that the Chinese government controls the decision making of foreign institutions. Therefore, if it is found that there is significant evidence of foreign institutional ownership, then it would imply that the main results are not driven by political connections or cultural similarities but related to the effect of institutional ownership per se.

[Insert [Table 9](#) Here]

[Table 9](#) re-estimates multinomial logistic regressions, but the focus here is on the sample firms with foreign institutional ownership. To identify whether a shareholder is foreign, the original sample data set is merged with the Orbis database maintained by Bureau van Dijk. Orbis database offers shareholder information regarding nationality, type⁴⁵, direct and total shareholdings.⁴⁶ For comparison purposes, [Table 9](#) conducts probit regressions with the merged foreign institutional ownership data for all cross-border listings and multinomial regressions with the merged data for each type of the cross-border listings. If firms with foreign institutional investors who emphasize the worldwide monitoring are significantly associated with overseas listings, then it would suggest that the findings documented above hold for the institutional ownership, netting of the influences of political connection.

The results are reported in [Table 9](#), where Panel A uses foreign institutional investors' shareholding (*Institutional Holding*) as a measure of institutional ownership. Panel A of [Table 9](#) suggests that the

⁴⁴ Bena et al. (2017) report that foreign institutions focus on the long-run strategic moves, including equity issuance and cross-border listings, that can exert a disciplinary role in entrenched corporate insiders. Further, Ferreira and Matos (2008) argue that foreign institutional investors in general have fewer potential ties with firm insiders and therefore are more independent. They find that foreign institutional investors have a strong preference for good governance.

⁴⁵ The Orbis data set classifies ownership into the following types: Bank, Corporate, Employees, Managers, Directors, Financial company, Foundation, Research Institute, Hedge fund, Insurance company, Mutual and pension fund, nominee, trust, trustee, One or more named individuals or families, Other unnamed shareholders, aggregated, Private equity firm, Public, Public authority, state, government, Self-ownership, Unnamed private shareholders, aggregated, Venture capital. A financial institution is defined as the shareholder belongs to the type of Bank, Financial company, Foundation, Hedge fund, Insurance company, Mutual and pension fund, nominee, trust, trustee, Private equity firm, and Venture capital.

⁴⁶ The issue with Orbis is that it includes shareholder information only starting from 2000, and in the early period, in particular in the period from 2000 through 2005, the information is largely missing. Thanks to the increased Chinese firms conducting foreign finance activities in recent years, with the post-2005 data, there is still sufficient information to carry out the tests for the influence of foreign institutional ownership despite a reduced sample size. There are around 400 for *Institutional Holding* regressions and around 500 for *Institutional Index* regressions.

shareholding percentage of the foreign institutional investor is positively related to the probability of all types of cross-border listings. Column (1) of Table 6 shows that a coefficient on *Institutional Holding* is 0.035 at the 1% significance level. The coefficient corresponds odds ratio of 1.067 from the logistic regression, suggesting that the odds of listing overseas are 6.7% higher with a 1% increase in foreign institutional holdings. Columns (2) to (6) demonstrate that the positive coefficient on *Institutional Holding* holds for all listing types, and the results are statistically significant at the 10% level or better. In addition, the effects of control variables are highly consistent with the results in previous regressions.

Panel B of [Table 9](#) uses the presence of foreign institutional shareholder(s) (*Institutional Index*) as a measure of institutional ownership. In Column (1), *Institution Index* has a coefficient of 1.640, the corresponding odds ratio from the logistic regression of 15.14. This suggests that for firms with foreign strategic institutional ownership, the odds of being listed overseas are fourteen times as large as the odds for firms without foreign institutional ownership. Columns (2) to (6) of Panel B show that *Institutional Index* is positively related to all types of cross-border listings at the 1% significance level. The coefficients on *Institution Index* for H-share listings, Red-chips listings, AH-listing, mainboard listings and GEM listings are generally more significant than those in Panel A where *Institutional Holding* is used as the measure of institutional ownership. These results suggest that having foreign institutional strategic investors on board significantly affects corporate decisions of listing overseas.

The results also show that the effect of foreign institutional ownership on H-share listings and the mainboard listings are highly and consistently positive and significant. This is in line with Doidge et al. (2009) who demonstrate that the main exchange listings are more likely to be affected by ownership structures. Unlike previous results obtained, however, the results here show that foreign institutional ownership is also important to GEM listings, suggesting that foreign institutional ownership is helpful for firms to list on the exchanges alternative to the main exchanges.

Overall, the results in this section suggest a stronger effect of institutional holdings compared to the earlier analyses where foreign institutions are not particularly considered. Therefore, the positive effect of institutional ownership on cross-border listings holds in the situation with less political influences.

4.2.3 Policy tool or privatization?

The above section provides evidence that the effect of institutional ownership on cross-border listings is not weakened by government and political influences. Given that it has been documented in the literature that government ownership plays a dominant role in cross-listings of Chinese firms, the follow-up question is whether institutional ownership has some effects on government control. Specifically, there are two explanations regarding the political consideration of cross-border listings of Chinese firms. One explanation upholds that cross-border listings simply serve the purpose of a policy tool (Hung et al., 2012; Sun et al., 2013). Based on this explanation, cross-border listings of government-controlled firms can be regarded as political achievements that are helpful for the political career and future promotion of managers. According to this argument, the government shareholdings would remain unchanged or even increase after the listing. Another argument advocates that the share issues can be an effective way of privatization (Boubakri et al., 2013; Sun & Tong, 2003), i.e., the government can sell their shares through public listings. According to this argument, government holdings are reduced after the listing. These competing arguments raise an interesting question: Whether the institutional holdings conspire with the government influence, or they are independent of political influence, or they can confine the political influence by optimizing the ownership structure as documented in Doidge et al. (2009).

To identify the firms that are controlled by the government, again Orbis dataset is employed. As mentioned earlier, Orbis gathers information regarding the type of shareholders. This allows not only to know whether a firm's ultimate owner is the government but also to attain the percentage of holdings of the controlling shareholders.⁴⁷ If institutional investors pushing a firm listing overseas is related to political reasons, the interaction between the institutional ownership and cross-border listings would be significant, either positive or negative.⁴⁸ Alternatively, if institutional investors propel cross-border

⁴⁷ The issue with the Orbis, again, is the data limitation. Fortunately, after filtering the data, there are still 906 observations. With this sample, the government ownership is regressed on institutional ownership (*Institutional Holding* or *Institutional Index*), cross-border listing types, the interaction between institutional ownership and each listing type, along with a set of control variables (*ROA*, *Sales growth*, *Firm size*, and *financial flexibility*). In each regression, the test-statistics are calculated based on standard errors clustered at the firm level.

⁴⁸ If institutional investors propel cross-border listings for political reasons, the cross-border listings should enhance government holdings. In contrast, if the privatization explanation holds, institutional investors play a

listings is not out of political reasons, the interaction between institutional ownership and cross-border listings are expected to be insignificant.⁴⁹

[Insert [Table 10](#) Here]

In [Table 10](#), Columns (1) to (4) show that the coefficients on *Institutional Holding* are significant in three out of four regressions (only except for the *Red-chip* regression), suggesting that institutional holdings indeed offset government ownership. Regarding the effect of cross-border listings on government ownership, the coefficient on *H-share listings* is significantly negative (in Column 2) and the coefficient on *Red-chips listing* is significantly positive (in Column 3). Therefore, the former provides support for the privatization argument and the latter supports the policy tool argument.⁵⁰ Further, Columns (5) to (8) of [Table 10](#) also show negative coefficients on *Institutional Index* in all regressions, suggesting that the existence of institutional investors can restrict government ownership. However, they are not statistically significant, indicating that holding shares instead of just presenting are more important for institutional investors to constrain government ownership. Besides, similar to the results in Columns (2) to (3), the coefficient on *H-share listing* in Column (6) is negative and on *Red-chips listing* in Column (7) is positive, these results seem to indicate that *H-share listings* are likely to serve as a way of the privatization, whereas *Red-chip listings* tend to be used as the policy tool.⁵¹

Overall, this subsection demonstrates that institutional strategic ownership, to some extent, restrains government control. Meanwhile, it provides evidence supporting both policy tool and privatization arguments. Therefore, cross-border listings advanced by institutional investors are not driven by political purposes. Taken together, the evidence indicates that institutional ownership facilitates

prominent role in the optimization of corporate governance, then the ownership structure is also likely to be optimized. In this case, government holdings tend to be reduced through the listings.

⁴⁹ Given that government-controlled firms are all listed on the mainboard of HKSE, it makes more sense to carry out this analysis according to the channels of listings, i.e. H-share listings, Red-chip listings, and AH-listings.

⁵⁰ When interacting *Institutional Holding* with listing types, the interaction terms are insignificant in all regressions, suggesting that all types of cross-border listings propelled by institutional investors are not out of political purpose.

⁵¹ Again, when interacting *Institutional Index* with listing types, there is no significant coefficient on the interaction terms between *Institutional Index* and listing type, suggesting that the purpose of institutional investors pushing a firm listing overseas is not related to politics.

optimizing ownership structures by promoting cross-border listings.

4.3 Institutional ownership and information environment

The bonding hypothesis suggests that cross-border listings bring positive constraints on firms, including direct constraints (such as legal enforcement) and indirect constraints (such as analyst coverage).⁵² In the previous section, the results show that the listings with stricter requirements (*H-share listings* and *Mainboard listings*) demonstrate a stronger effect, this provides support for the effect of direct constraints. This section tests indirect constraints. Doidge et al. (2009) document that greater analyst coverage increases market scrutiny and monitoring, which is strongly related to a better corporate information environment. For Chinese firms, a higher level of analyst coverage improves firm-specific information disclosure and pricing efficiency, which have been documented to benefit cross-border listed Chinese firms (Chan et al., 2008; Chang, Luo, & Ren, 2013; Li et al., 2015; Tourani-Rad, Gilbert, & Chen, 2016). Therefore, if institutional ownership plays a role in affecting indirect constraints, then there would be a significant relation between institutional ownership and firm information environment.

[Insert [Table11](#) Here]

To test whether institutional ownership improves the information environment of cross-border listed firms, [Table11](#) follows the approach of Doidge et al. (2009) and construct *Analyst coverage* as the number of analysts. Also similar to Doidge et al. (2009), we constructs two control variables: (1) *Earning surprise*, measured by the annual absolute value of the difference between actual and forecasted earnings per shares scaled by the mean of the forecasted earnings per shares, and (2) *Forecast dispersion*, measured by the annual standard deviation of the forecasted earnings per shares, scaled by the mean of the forecasted earnings per shares. The data used for constructing these two variables are obtained from Datastream. [Table11](#) estimates the specifications with standard errors clustered at the firm-level.⁵³ Specifically, it regresses *Analyst coverage* separately on institutional ownership and each type of the

⁵² Doidge et al. (2009) find that the benefit of cross-border listing associated with the ownership structure mainly comes from the direct constraints, i.e. legal enforcement, of the developed exchanges.

⁵³ Similar to the ones used by Doidge et al. (2009), all these specifications control for *Earning surprise*, *Forecast dispersion*, and *Firm size*.

cross-border listings.⁵⁴ If institutional ownership helps improve the corporate information environment, then *Analyst coverage* would be related to institutional strategic ownership.

In [Table 11](#), Panel A reports results from the regressions of the analyst coverage on *Institutional Holding* and listing types. In Column (1), where the listing effect is not controlled, the coefficient on *Institutional Holding* is positive and significant at the 1% level, suggesting that *Institutional Holding* positively affects the corporate information environment. Throughout the regressions regarding cross-border listings (Columns 2 to 5), all the coefficients on *Institutional Holding* remain positive and significant at the 5% level or better. Further, cross-border listings also show a positive effect on the corporate information environment and significant in three out of four regressions (only except for *H-share listings* in Column 2). However, when testing domestic listings (Column 6), *Institutional Holding* does not demonstrate the significant effect, and *A-share listing* is negatively related to analyst coverage. These results imply that cross-border listings improve the firm information environment, but domestic listings do not. Subsequently, the positive effect of institutional ownership is mainly manifested in cross-border listings. Prior studies suggest that both institutional ownership and cross-border listing facilitate information disclosure and improve firm information environment (e.g., Bird & Karolyi, 2016; Fernandes & Ferreira, 2008). The evidence documented in Column (3) is in line with this argument.

Panel B of [Table 11](#) reports the results from regressions with *Institutional Index*. Similar to the above results from *Institutional Holding*, the coefficients on *Institutional Index* are consistently positive and statistically significant, even for the domestic listing regression. Nonetheless, the statistical significance is higher for *Institutional Index* than for *Institutional Holding*. These results confirm the importance of having institutional investors in improving the corporate information environment. Same to the results in Panel A, Panel B shows that cross-border listings positively affect the corporate information environment (Columns 2 to 5) and *A-share listing* negatively affects the corporate information environment.⁵⁵ In addition, throughout Columns (1) to (6) in both panels, *Firm size* is positive and

⁵⁴. Except for *GEM listings* as *GEM listings* do not offer sufficient data on *Analyst coverage* to carry out the regression analysis.

⁵⁵ When interacting institutional ownership and listing types, the results show that the interaction between *Institutional Index* and *A-share listing* is significantly negative, suggesting that the positive effect of institutional

Forecast dispersion is negatively associated with analyst coverage. These results are consistent with Doidge et al. (2009). Taken together, the results in previous sections of this paper show that firms with institutional ownership are more likely to pursue a cross-border listing. The results in this section further reveal that the positive effect of institutional ownership is associated with the information environment.

Overall, the results in this section confirm the indirect constraint argument of cross-border listings and indicate that institutional investors play a positive role in improving corporate information in different types of cross-border listings, though listing domestically attenuates this effect. Therefore, institutional ownership facilitates the improvement of the information environment of emerging market firms.

4.4 Ex-post determinants of institutional ownership

The previous sections show that institutional ownership facilitates cross-border listings, which in turn optimize the ownership structure and improve the information environment of emerging market firms. This naturally leads to a follow-up question: With what characteristics, a firm can attract institutional investors? This section deals with this question using ex-post analyses.⁵⁶ To examine the determinants of institutional ownership, this section follows the literature and use Cox's (1972) proportional hazard model literature (Ayyagari & Doidge, 2010; Doidge et al., 2009). Cox model predicts factor changes in a dynamic way, which allow predicting those⁵⁷ firms that do not have institutional investors in year t are more likely to have institutional investors in year $t+1$. In other words, Cox model allows estimating the probability of the appearance of institutional ownership in year $t+1$ as a function of a range of variables.⁵⁸ The model can be written as,

$$H(T|X_j) = H_0(T)EXP(X_j\beta_X) \quad (2)$$

ownership on analyst coverage could be largely offset if the firm is listed domestically. On the other hand, the positive coefficients on interactions between *Institutional Index* and *Cross-border listing types* suggest that the positive effect of institutional ownership on analyst coverage is enhanced if cross-border listings.

⁵⁶ Because corporate financial data before listing are rarely publicly available and are hence very limited, it is unlikely to perform a sound ex-ante analysis. Fortunately, publicly available financial information after the listing allows conducting ex-post analyses.

⁵⁷

⁵⁸ For a detailed discussion of the Cox model, including its advantages and disadvantages, see, for example, Ayyagari and Doidge (2010) and Doidge et al. (2009).

where $H(T|X_j)$ is the hazard rate for firm j . In this study, the hazard rate is the probability of a firm to have institutional investor, given that it has not yet had one in prior years.⁵⁹ X_j is a range of independent variables. $H_0(T)$ is the baseline hazard for firm j corresponds to the hazard rate with X_j set to zero. β_X is a vector of coefficients to be estimated. $EXP(X_j\beta_X)$ as a whole is the relative hazard and is the function of independent variables. As expressed, the hazard rate is obtained by shifting the baseline hazard as the independent variables change. Thus, the model assumes that no matter what shape of the baseline hazard, it is the same for all firms. In all regressions, standard errors clustered at the firm level.

[Insert [Table 12](#) Here]

The model is estimated three times in [Table 12](#). It begins with including all firms, i.e., both China incorporated firms and overseas incorporated firms as well as domestically and cross-border listed firms. Column (1) of [Table 12](#) reports this result. It shows that the hazard ratio is significant at the 10% level or better in five factors: *Firm size*, *Financial Flexibility*, *Difference in HDI*, *Log GDP*, and *Analyst coverage*. Among them, the coefficients on *Firm size*, *Difference in HDI*, and *Log GDP* are negative. While coefficients on *Financial flexibility* and *Analyst coverage* are positive. These results suggest that in general, institutional investors tend to seek firms with smaller size and, meanwhile, these firms should have higher levels of information disclosure. As for regional characteristics, institutional strategic investors focus on the areas with better social developments, as suggested by the negative effect of *Difference in HDI* (the smaller the difference, the closer to the level of Hong Kong), instead of the areas with economic advantages, as indicated by the negative coefficient on the *Log GDP*.

Column (2) of [Table 12](#) reports the regression results of China incorporated firms. The result shows that four of the five factors documented in Column (1) of [Table 12](#) remain significant, they are *Firm size*, *Financial Flexibility*, *Difference in HDI*, and *Analyst coverage*, and their signs are consistent with the above result. These results suggest that for China incorporated firms, the institutional strategic investors

⁵⁹ To be specific, the dependent variable equals one in the year when an institutional investor appears as a strategic shareholder of firm j . After that year, firm j is dropped from the hazard model.

focus on small firms and emphasize on better firm quality (a higher level of financial flexibility and information disclosure) as well as regional social development (a higher level of human development). This finding is consistent with Doidge et al. (2009), indicating that for the Chinese mainland incorporated firms, institutional investors are more likely to have ownership in the firms with a better corporate environment of information disclosure and transparency.

Column (3) of [Table 12](#) reports results from a sample of China incorporated firms that are also cross-border listed. The results obtained here show some differences from those in Column (1) and (2): For those firms that have already been cross-border listed, institutional strategic investors seem to emphasize more on the regional social development instead of firm-level characteristics, as suggested by the significant coefficient on *Log GDP* and *Difference in HDI*. The evidence in this section overall suggests that smaller firms are more likely to attract institutional strategic investors if the firms have a higher level of financial flexibility, with a transparent information environment, and incorporated in the regions with better social developments. The focus of institutional investors, therefore, is not the scale of the company, but the developmental potential, financial health, and informational transparency.

5 Conclusion

Recent studies suggest that issuing shares in developed markets significantly benefits firms from emerging markets, because such cross-border listings can bond the emerging market firms to better legal institutions in the developed markets and thus help improve their corporate governance (Karolyi, 2004, 2012). However, emerging market firms may be reluctant to increase information disclosure, as requested by the developed market exchanges. Because increasing information disclosure may affect the private benefits of the insiders of these firms. As a result, emerging market firms tend to be unwilling to list overseas, even though they know there could be potential benefits by doing so.

This study posits that institutional investors play an important role in cross-listings of emerging market firms. Using a sample of Hong Kong-listed Chinese firms, it shows that institutional strategic ownership is positively related to the probability of the cross-border listings, after controlling for the firm- and

regional-level characteristics as well as industry-level shocks. The positive effect of institutional investors on cross-border listings holds consistently after addressing endogeneity and implementing exogenous shocks. Further, the evidence across geographical regions and from foreign institutional investors shows that the effect of institutional ownership is not attenuated by political influence and, to some extent, constrains government shareholdings. The results also reveal that institutional investors play a positive role in improving corporate information in cross-border listings and thus confirm indirect constraint of cross-border listings. Finally, in modeling the features underlying a firm to attract strategic institutional investors, we find that institutional ownership is positively related to the level of financial flexibility, the quality of the corporate information environment and the regional social development.

Overall, this study connects the literature on the strategic role of institutional investors and international listings and offers novel evidence on emerging markets. The results of this paper extend the existing literature and show that by facilitating cross-border listings, institutional ownership plays a significant role in helping integrate emerging market firms in the world economy.

References

- Arquette, G. C., Brown, W. O., & Burdekin, R. C. K. (2008). US ADR and Hong Kong H-share discounts of Shanghai-listed firms. *Journal of Banking and Finance*, 32(9), 1916–1927.
- Ayyagari, M., & Doidge, C. (2010). Does cross-listing facilitate changes in corporate ownership and control? *Journal of Banking and Finance*, 34(1), 208–223.
- Baik, B., Kang, J.-K., & Kim, J.-M. (2010). Local institutional investors, information asymmetries, and equity returns. *Journal of Financial Economics*, 97(1), 81–106.
- Bekaert, G. (1995). Market integration and investment barriers in emerging equity markets. *World Bank Economic Review*, 9(1), 75–107.
- Bena, J., Ferreira, M. A., Matos, P., & Pires, P. (2017). Are foreign investors locusts? The long-term effects of foreign institutional ownership. *Journal of Financial Economics*, 126(1), 122–146.
- Bird, A., & Karolyi, S. A. (2016). Do institutional investors demand public disclosure? *Review of Financial Studies*, 29(12), 3245–3277.
- Boubakri, N., Cosset, J.-C., & Saffar, W. (2013). The role of state and foreign owners in corporate risk-taking: Evidence from privatization. *Journal of Financial Economics*, 108(3), 641–658.
- Busaba, W. Y., Guo, L., Sun, Z., & Yu, T. (2015). The dark side of cross-listing: A new perspective from China. *Journal of Banking and Finance*, 57, 1–16.
- Chan, K., Menkveld, A., & Yang, Z. (2008). Information asymmetry and asset prices: Evidence from the China foreign share discount. *Journal of Finance*, 63(1), 159–196.
- Chang, E. C., Luo, Y., & Ren, J. (2013). Cross-listing and pricing efficiency: The informational and anchoring role played by the reference price. *Journal of Banking and Finance*, 37(11), 4449–4464.
- Chemmanur, T. J., & He, J. (2011). IPO waves, product market competition, and the going public decision: Theory and evidence. *Journal of Financial Economics*, 101(2), 382–412. <https://doi.org/https://doi.org/10.1016/j.jfineco.2011.03.009>
- Chemmanur, T. J., He, S., & Hu, G. (2009). The role of institutional investors in seasoned equity offerings. *Journal of Financial Economics*, 94(3), 384–411.
- Chemmanur, T. J., Hu, G., & Huang, J. (2010). The role of institutional investors in initial public offerings. *Review of Financial Studies*, 23(12), 4496–4540.
- Chemmanur, T. J., & Liu, M. H. (2011). Institutional trading, information production, and the choice between spin-offs, carve-outs, and tracking stock issues. *Journal of Corporate Finance*, 17(1), 62–82.
- Chiang, Y.-M., Qian, Y., & Sherman, A. E. (2009). Endogenous entry and partial adjustment in IPO auctions: Are institutional investors better informed? *Review of Financial Studies*, 23(3), 1200–1230.
- Chung, K. H., Elder, J., & Kim, J.-C. (2010). Corporate governance and liquidity. *Journal of Financial and Quantitative Analysis*, 45(2), 265–291.
- Claessens, S., Djankov, S., & Lang, L. H. P. (2000). The separation of ownership and control in East Asian Corporations. *Journal of Financial Economics*, 58(1), 81–112.
- Claessens, S., Klingebiel, D., & Schmukler, S. (2002). *Explaining the migration of stocks from exchanges in emerging economies to international centers* (WPS2816). The World Bank.
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society. Series B (Methodological)*, 34(2), 187–220.
- Dahlquist, M., & Robertsson, G. (2001). Direct foreign ownership, institutional investors, and firm characteristics. *Journal of Financial Economics*, 59(3), 413–440.
- Doidge, C., Karolyi, G. A., Lins, K. V., Miller, D. P., & Stulz, R. M. (2009). Private benefits of control, ownership, and the cross-listing decision. *Journal of Finance*, 64(1), 425–466.
- Doidge, C., Karolyi, G. A., & Stulz, R. M. (2004). Why are foreign firms listed in the US worth more? *Journal of Financial Economics*, 71(2), 205–238.
- Doidge, C., Karolyi, G. A., & Stulz, R. M. (2007). Why do countries matter so much for corporate governance? *Journal of Financial Economics*, 86(1), 1–39.
- Domowitz, I., Glen, J., & Madhavan, A. (1998). International cross-listing and order flow migration: Evidence from an emerging market. *Journal of Finance*, 53(6), 2001–2027.
- Edison, H. J., & Warnock, F. E. (2004). U.S. investors' emerging market equity portfolios: A security-

- level analysis. *Review of Economics and Statistics*, 86(3), 691–704.
- Farrar, D. E., & Gorton, L. (1981). Institutional investors and concentration of financial power: Berle and Means revisited. *Journal of Finance*, 36(2), 369–381.
- Fernandes, N., & Ferreira, M. A. (2008). Does international cross-listing improve the information environment. *Journal of Financial Economics*, 88(2), 216–244.
- Ferreira, M. A., Massa, M., & Matos, P. (2009). Shareholders at the gate? Institutional investors and cross-border mergers and acquisitions. *Review of Financial Studies*, 23(2), 601–644.
- Ferreira, M. A., & Matos, P. (2008). The colors of investors' money: The role of institutional investors around the world. *Journal of Financial Economics*, 88(3), 499–533.
- Gibson, M. S. (2003). Is corporate governance ineffective in emerging markets? *The Journal of Financial and Quantitative Analysis*, 38(1), 231–250.
- Goto, S., Watanabe, M., & Xu, Y. (2008). Strategic disclosure and stock returns: Theory and evidence from US cross-listing. *Review of Financial Studies*, 22(4), 1585–1620.
- Hung, M., Wong, T. J., & Zhang, T. (2012). Political considerations in the decision of Chinese SOEs to list in Hong Kong. *Journal of Accounting and Economics*, 53(1), 435–449.
- Jenkinson, T., & Jones, H. (2008). IPO pricing and allocation: a survey of the views of institutional investors. *Review of Financial Studies*, 22(4), 1477–1504.
- Jiang, F., & Kim, K. A. (2015). Corporate governance in China: A modern perspective. *Journal Of Corporate Finance*, 32, 190–216.
- Karolyi, G. A. (2004). The role of American depositary receipts in the development of emerging equity markets. *Review of Economics and Statistics*, 86(3), 670–690.
- Karolyi, G. A. (2006). The world of cross-listings and cross-listings of the world: Challenging conventional wisdom. *Review of Finance*, 10(1), 99–152.
- Karolyi, G. A. (2012). Corporate governance, agency problems and international cross-listings: A defense of the bonding hypothesis. *Emerging Markets Review*, 13(4), 516–547.
- King, M. R., & Segal, D. (2009). The long-term effects of cross-listing, investor recognition, and ownership structure on valuation. *Review of Financial Studies*, 22(6), 2393–2421.
- Lang, L. H. P. (2005). *Governance and expropriation*. Edward Elgar Publishing.
- Lemmon, M. L., & Lins, K. V. (2003). Ownership structure, corporate governance, and firm value: Evidence from the East Asian financial crisis. *Journal of Finance*, 58(4), 1445–1468.
- Li, S., Brockman, P., & Zurbuegg, R. (2015). Cross-listing, firm-specific information, and corporate governance: Evidence from Chinese A-shares and H-shares. *Journal of Corporate Finance*, 32, 347–362.
- Lins, K. V. (2003). Equity ownership and firm value in emerging markets. *Journal of Financial and Quantitative Analysis*, 38(1), 159–184.
- Lins, K. V., Strickland, D., & Zenner, M. (2005). Do non-US firms issue equity on US stock exchanges to relax capital constraints? *Journal of Financial and Quantitative Analysis*, 40(01), 109–133.
- Pagano, M., Röell, A. A., & Zechner, J. (2002). The geography of equity listing: why do companies list abroad? *Journal of Finance*, 57(6), 2651–2694.
- Sarkissian, S., & Schill, M. J. (2004). The overseas listing decision: New evidence of proximity preference. *Review of Financial Studies*, 17(3), 769–809.
- Stock, J. H., Wright, J. H., & Yogo, M. (2002). A survey of weak instruments and weak identification in generalized method of moments. *Journal of Business & Economic Statistics*, 20(4), 518–529.
- Stulz, R. M. (2005). The limits of financial globalization. *Journal of Finance*, 60(4), 1595–1638.
- Sun, Q., & Tong, W. H. S. (2003). China share issue privatization: the extent of its success. *Journal of Financial Economics*, 70(2), 183–222.
- Sun, Q., Tong, W. H. S., & Wu, Y. (2013). Overseas listing as a policy tool: Evidence from China's H-shares. *Journal of Banking and Finance*, 37(5), 1460–1474.
- Sundaram, A. K., & Logue, D. E. (1996). Valuation effects of foreign company listings on U.S. exchanges. *Journal of International Business Studies*, 27(1), 67–88.
- Tourani-Rad, A., Gilbert, A., & Chen, J. (2016). Are foreign IPOs really foreign? Price efficiency and information asymmetry of Chinese foreign IPOs. *Journal of Banking and Finance*, 63, 95–106.

Table 1

Sample distribution. This table presents a distribution of the sample of 2,603 Chinese firms listed domestically and on the Hong Kong Stock Exchange from 1990 to 2017. Panels A, B and C reports the sample by year, industry and region, respectively.

	Domestic listed	Hong Kong-listed					AH-listed
	A-shares	All	H-shares	Red-chips	Mainboard	GEM	
Whole sample	2,246	259	146	113	231	28	98
Panel A: By years							
1990-1995	242	13	3	10	13	0	19
1996-2000	332	38	7	31	36	2	25
2001-2005	208	64	41	23	45	19	21
2006-2010	47	41	20	21	38	3	23
2011-2015	692	75	52	23	73	2	10
Post 2015	725	28	23	5	26	2	0
Panel B: By industry							
Basic Materials	232	15	7	8	13	2	9
Communications	122	14	7	7	11	3	1
Consumer, Cyclical	401	31	14	17	26	5	12
Consumer, Non-cyclical	349	39	22	17	34	5	12
Diversified	18	1	1	0	0	0	0
Energy	70	11	7	4	11	0	7
Financial	131	60	34	26	60	2	25
Industrial	719	56	36	20	49	5	28
Technology	148	16	11	5	10	6	1
Utilities	56	15	7	8	15	0	3
Panel C: By region of headquarter							
Overseas	0	98	1	97	92	6	0
Beijing	340	55	50	5	52	3	34
Guangdong	408	18	16	2	15	3	19
East	926	52	45	7	41	11	27
Middle and Northeast	294	17	16	1	14	3	11
West	278	19	18	1	17	2	7

Table 2

Summary statistics for regression variables. This table reports summary statistics for the key regression variables used in the hypotheses tests to examine the impact of institutional ownership on cross-border listings of the sample firms. Institutional holding is the percentage of shares held by institutional investors. Institutional index is a dummy variable that equals one if a (or more) strategic institutional investor appears in a sample firm, and otherwise zero. ROA is return on assets. Sales growth is the annual percentage change in sales. Firm size is the natural logarithm of total assets. Leverage is the ratio of total debt to total capital. Offer price is the stock price initial offer price in Hong Kong dollars. Offer Size is the total amount of fund raised through public offering in Hong Kong dollars. Financial flexibility is an index ranging from zero to three, where values increase by one point if the firm is at the position above the 75th percentiles of cash ratio, by one point if the firm is at the position above the 75th percentiles of dividend ratio, and by one point if the firm is at the position below the 25th percentiles of the capital expenditure to total assets ratio. Difference in HDI is the difference in the human development index between Hong Kong and the province for a given firm. Log foreign trade is the natural logarithm of the foreign trading volume of the province for a given firm. Log GDP is the natural logarithm of GDP per capita of the province for a given firm. Analyst coverage is the annual count of analyst associated with a firm after listing. Earning surprise is the annual absolute value of difference between actual and forecasted earning per shares, scaled by the mean of the forecasted earning per shares. Forecast dispersion is the annual standard deviation of the forecasted earning per shares, scaled by the mean of the forecasted earning per shares. Market to book value is the post-listing market to book value of equity. Government holding is the percentage of post-listing holdings by the largest shareholder, where government is the global ultimate owner. The detailed definitions and data sources for the regression variables are summarized in Appendix 1.

	N	Mean	Standard deviation	Minimum	Median	Maximum
Institutional Holding (%)	2,544	0.72	2.65	0.00	0.00	54.67
Institutional Index	2,544	0.21	0.41	0.00	0.00	1.00
ROA (%)	1,922	9.69	14.70	-98.31	8.68	605.02
Sales growth (%)	1,923	20.18	36.06	-98.87	15.33	854.58
Firm size	2,048	21.30	1.47	13.80	20.90	29.96
Leverage	2,047	16.72	18.20	0.00	10.89	105.92
Offer Price (dollors)	2,602	11.26	10.86	0.00	8.85	113.20
Offer Size (billion dollares)	2,602	0.65	4.31	0.00	0.06	129.51
Financial flexibility	2,602	1.24	0.84	0.00	1.00	3.00
Log foreign trade	2,500	22.84	1.61	17.31	23.36	24.69
Log GDP	2,500	11.30	0.38	10.29	11.33	11.77
Difference in HDI	2,500	0.12	0.05	0.04	0.14	0.31
Analyst coverage	9,346	6.05	6.57	1.00	4.00	38.00
Earning surprise	9,335	0.35	1.08	0.00	0.15	54.22
Forecast dispersion	7,240	0.18	1.28	0.00	0.09	93.91
Market to book value	2,053	4.66	3.65	0.29	3.71	45.67
Government ownership (%)	969	43.40	23.19	0.00	42.96	100.00

Table 3

Cross-sectional multinomial logistic regressions: Institutional investors and the choice of the type of cross-border listing. This table reports coefficients on the strategic shareholdings of institutional investors with a set of control variables regarding the types of cross-border listings of Chinese firms. The baseline group is domestically listed Chinese firms. The types of cross-border listings are classified by two ways. The first way is based on the channel through which a Chinese firm list to the Hong Kong Stock Exchange, including H-shares, Red-chips and AH-listings. The second way is based on the listing venues of the Hong Kong Stock Exchange, including the mainboard and the Growth Enterprise Market (GEM). Institutional Holding is the percentage of strategic free float shares held by institutional investors. Institution Index is a dummy variable that equals one if a firm has strategic institutional shareholder(s). The definitions and data sources for control variables are detailed in Appendix 1. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. Pseudo- R^2 is a goodness-of-fit measure according to the difference between restricted and unrestricted likelihood functions. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Panel A					Panel B				
	H-shares	Red-chips	AH-listings	Mainboard	GEM	H-shares	Red-chips	AH-listings	Mainboard	GEM
Institutional Holding	0.259*** (3.69)	0.069 (0.58)	0.230*** (3.83)	0.260*** (3.58)	0.458*** (3.51)					
Institutional Index						1.663*** (7.02)	0.762 (1.17)	2.332*** (6.88)	2.060*** (7.92)	-2.429 (-1.54)
ROA	-0.003 (-0.11)	0.010*** (3.01)	0.004* (1.68)	0.003 (0.54)	0.010 (0.20)	0.000 (0.01)	0.011*** (3.17)	0.006** (2.24)	0.004 (1.00)	0.013 (0.93)
Sales growth	0.005* (1.66)	0.003 (0.43)	0.008** (2.20)	0.005 (1.63)	0.021*** (3.13)	0.004 (1.17)	0.003 (0.48)	0.006 (1.63)	0.003 (1.04)	0.018*** (3.87)
Firm size	0.401*** (4.01)	0.692*** (2.80)	1.023*** (7.46)	0.675*** (5.84)	-2.408** (-2.46)	0.498*** (4.83)	0.744*** (2.89)	1.127*** (8.31)	0.789*** (6.86)	-1.694*** (-3.02)
Leverage	0.003 (0.47)	-0.017 (-1.19)	-0.032*** (-3.50)	-0.005 (-0.59)	0.035 (0.69)	-0.001 (-0.08)	-0.019 (-1.36)	-0.037*** (-4.04)	-0.01 (-1.17)	0.035 (0.96)
Financial flexibility	0.608*** (4.00)	0.959** (2.35)	0.029 (0.14)	0.579*** (3.62)	1.133 (1.46)	0.510*** (3.23)	0.926** (2.26)	-0.095 (-0.46)	0.454*** (2.74)	1.101 (1.28)
Offer price	-0.193*** (-2.97)	-0.314*** (-4.23)	-0.164*** (-3.31)	-0.151*** (-2.82)	-1.243*** (-2.91)	-0.189*** (-2.94)	-0.314*** (-4.10)	-0.153*** (-3.79)	-0.146*** (-2.96)	-0.955*** (-4.29)
Difference in HDI	-4.408 (-0.80)	7.946 (0.50)	-5.809 (-0.68)	-1.123 (-0.20)	17.785 (0.65)	-1.663 (-0.31)	10.595 (0.64)	-1.004 (-0.12)	2.700 (0.47)	12.87 (0.76)
Log foreign trade	-0.147 (-1.23)	-0.124 (-0.47)	0.164 (0.87)	-0.114 (-0.94)	-1.747*** (-4.15)	-0.141 (-1.23)	-0.134 (-0.53)	0.197 (1.03)	-0.107 (-0.90)	-0.941 (-1.45)
Log GDP	0.616 (0.77)	3.182 (1.28)	-1.075 (-0.77)	0.915 (1.07)	9.412** (2.04)	0.933 (1.16)	3.48 (1.39)	-0.644 (-0.44)	1.328 (1.50)	5.631* (1.82)
Constant	-14.587 (-1.63)	-52.547* (-1.68)	-15.079 (-0.98)	-25.117*** (-2.67)	-25.487 (-0.59)	-20.724** (-2.31)	-57.438* (-1.79)	-24.12 (-1.53)	-33.061*** (-3.37)	-13.81 (-0.46)
Industry-dummy	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.412	0.412	0.412	0.479	0.479	0.417	0.417	0.417	0.495	0.495
N of Observation	1,842	1,842	1,842	1,842	1,842	1,842	1,842	1,842	1,842	1,842

Table 4

Two-stage least square (2SLS) Instrumental variables approach. This table reports instrumental variables regression estimation results. Offer Size is the instrumental variable for Institutional Holding and Institutional Index and is measured by the total amount of fund raised through public offering in Hong Kong dollars. Column 1 reports the first-stage results of the 2SLS regressions with *Institutional Holding* as the dependent variable. Institutional holding is the percentage of shares held by institutional investors. Column 3 reports the first-stage results of the 2SLS regressions with *Institutional index* as the dependent variable. Institutional index is a dummy variable that equals one if a (or more) strategic institutional investor appears in a sample firm, and otherwise zero. Columns 2 and 4 report the second-stage results of the 2SLS regressions with *Cross-border listing* as the dependent variable. All other variables are defined in Appendix A. and 4 report the second-stage results from 2SLS regressions for Cross-border listing. Standard errors are clustered at the firm level and t-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Dependent variable	Institutional Holding	Cross-border listing	Institutional Index	Cross-border listing
	First stage	Second stage	First stage	Second stage
	(1)	(2)	(3)	(4)
Institutional Holding		0.393*** (7.68)		
Institutional Index				3.012*** (42.92)
Offer size	1.0501*** (5.75)		0.125*** (6.36)	
ROA	-0.000 (-0.25)	-0.002 (-0.54)	-0.001*** (-2.71)	0.001 (0.17)
Sales growth	0.003 (0.84)	-0.001 (-0.68)	0.000 (1.29)	-0.000 (-1.05)
Firm size	-0.648*** (-3.89)	0.002 (0.04)	-0.067*** (-3.38)	-0.030 (-0.82)
Leverage	0.021*** (2.90)	-0.004* (-1.70)	0.003*** (3.41)	-0.004* (-1.76)
Financial flexibility	0.035 (0.38)	-0.000 (-0.00)	0.009 (0.77)	-0.014 (-0.37)
Offer price	-0.019*** (-3.01)	-0.003 (-0.92)	-0.001 (-1.40)	-0.006 (-1.54)
Difference in HDI	-3.956 (-1.51)	2.293* (1.92)	-0.946** (-2.06)	3.481** (2.25)
Log foreign trade	0.072 (1.23)	-0.021 (-0.75)	0.017** (2.12)	-0.045 (-1.59)
Log GDP	-0.453 (-1.30)	0.291 (1.44)	-0.152** (-2.09)	0.555** (2.17)
Constant	20.871*** (3.82)	-3.861* (-1.67)	3.209*** (3.56)	-5.641* (-1.93)
Industry-dummy	YES	YES	YES	YES
Wald test of exogeneity		129.49 [<0.001]		222.99 [<0.001]
F-statistic	7.32 [<0.001]		12.05 [<0.001]	
Wald χ^2		317.96 [<0.001]		2152.40 [<0.001]
Adjusted R^2	0.122		0.195	
Partial R^2 of the instrument	0.045		0.032	
N of Observation	1,821	1,821	1,807	1,807

Table 5

Propensity score matching. This table reports tests results from propensity score-matching techniques. Panel A of Table 5 shows the summary statistics of treated observations (firms with institutional investors) and controlled observations (firms without institutional investors). Panel B of Table 5 presents the results from the first-stage of propensity score matching tests, i.e. the probit model. The dependent variable is the dummy variable of treated observations, that is, whether or not the firm has institutional investors. This stage generates propensity scores that will be used for matching in the second stage. Panel C of Table 5 reports average treatment effect of treated (ATET) with four propensity score-matching techniques: ATET Nearest-neighbor, ATET Radius, ATET Kernel, ATET Stratification. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Columns	(1)	(2)
Panel A: Summary statistics		
Treatment	Number of observations	Percent frequency
0	2,020	79.40%
1	524	20.60%
Panel B: First stage probit coefficients		
	Coefficients	(z-statistics)
ROA	-0.007	(-0.97)
Sales growth	0.003***	(2.83)
Firm size	0.081***	(2.97)
Leverage	0.005**	(2.23)
Financial flexibility	0.212***	(4.86)
Offer price	-0.011***	(-2.97)
Difference in HDI	-7.375***	(-3.74)
Log foreign trade	-0.006	(-0.15)
Log GDP	-0.805**	(-2.51)
Panel C: Second-stage Propensity score matching		
	Coefficients	(t-statistics)
ATET Nearest-neighbor	0.199***	(8.61)
ATET Radius	0.307***	(13.40)
ATET Kernel	0.224***	(18.31)
ATET Stratification	0.210***	(14.29)

Table 6

The exogenous shocks on the effect of institutional investors on cross-border listings. This table reports results from probit regressions that estimate the passage of regulations on the effect of institutional ownership on cross-border listings. The dependent variable in each regression is a dummy variable that equals one for cross-border listing and zero otherwise. Shocks are implemented as dummy variables of Post-1997, post-2004, and Post-2012 when the passage of “Circular of the State Council Concerning Further Strengthening the Administration of Share Issuance and Listing Overseas”, “Circular of the China Securities Regulatory Commission on Several Issues Concerning the Standardization on Overseas Listing of Subordinated Enterprises of Domestically-Listed Companies” and Several Opinions on Further Promoting the Healthy Development of the Capital Market”, and “Special Provisions of the State Council Concerning the Floatation and Listing Abroad of Stocks by Limited Stock Companies”, respectively. The key variable of interest is institutional investors’ shareholding, measured by Institutional Holding that is the percentage of strategic free float shares held by foreign institutional investors and by Institutional Index that is a dummy variable that equals one if a firm has strategic foreign institutional shareholder(s). Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The definitions and data sources for control variables are detailed in Appendix 1. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable	Cross-border listing					
	(1)	(2)	(3)	(4)	(5)	(6)
Institutional Holding	0.083** (1.99)	0.071* (1.66)	0.048 (1.19)			
Institutional Index				0.929*** (8.49)	1.287*** (8.92)	1.336*** (6.67)
Post-1997 dummy	-0.430 (-0.66)			0.936 (1.52)		
Post-2004 dummy		0.1295 (0.87)			-0.475** (-2.47)	
Post-2012 dummy			0.170 (1.48)			-0.249 (-1.28)
Post-1997 * Institutional Holding	4.852** (2.15)					
Post-2004 * Institutional Holding		0.140** (2.31)				
Post-2012 * Institutional Holding			0.161*** (3.17)			
Post-1997 * Institutional Index				-1.150 (-1.35)		
Post-2004 * Institutional Index					0.929*** (4.06)	
Post-2012 * Institutional Index						0.680*** (2.80)
ROA	0.001 (0.56)	0.001 (0.46)	0.001 (0.66)	0.002 (0.94)	0.001 (0.68)	0.002 (0.76)
Sales growth	0.003** (2.38)	0.003** (2.23)	0.003** (2.26)	0.003* (1.81)	0.003** (2.01)	0.003* (1.83)
Firm size	0.345*** (7.10)	0.356*** (7.44)	0.325*** (7.18)	0.350*** (7.39)	0.357*** (7.30)	0.358*** (7.68)
Leverage	-0.004 (-1.20)	-0.004 (-1.15)	-0.002 (-0.63)	-0.004 (-1.18)	-0.005 (-1.28)	-0.004 (-1.09)
Financial flexibility	0.265*** (4.30)	0.251*** (3.89)	0.256*** (3.73)	0.229*** (3.45)	0.204*** (2.96)	0.199*** (2.85)
Offer price	-0.074*** (-4.19)	-0.070*** (-3.92)	-0.068*** (-3.93)	-0.074*** (-4.17)	-0.075*** (-4.06)	-0.074*** (-4.33)
Difference in HDI	-2.863 (-1.26)	-2.393 (-1.03)	-2.056 (-0.87)	-1.694 (-0.70)	-1.257 (-0.51)	-0.885 (-0.36)
Log foreign trade	-0.040 (-0.80)	-0.040 (-0.80)	-0.044 (-0.89)	-0.054 (-1.05)	-0.055 (-1.03)	-0.046 (-0.86)
Log GDP	0.113 (0.32)	0.204 (0.55)	0.256 (0.69)	0.325 (0.85)	0.382 (0.97)	0.411 (1.04)
Constant	-8.6019** (-2.17)	-10.0739** (-2.41)	-10.1363** (-2.44)	-11.1249*** (-2.59)	-12.0693*** (-2.71)	-12.7761*** (-2.92)
Industry-dummy	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.445	0.450	0.466	0.461	0.472	0.468
N of Observation	1,842	1,842	1,842	1,842	1,842	1,842

Table 7

Institutional holdings by region. This table reports the average percentage of shares holdings by institutional investors. The shareholding is defined as the free float strategic holdings by Datastream. The statistics for ownership structure of sample firms are computed as of December 31, 2017 for Chinese firms that are listed domestically, on the Hong Kong stock exchange purely, and AH-listed in both Chinese and Hong Kong markets. For Hong Kong-listed firms, the statistics is also given in terms of listing channel — H-shares and Red-chips, and listing venues — Mainboard and Growth Enterprise Market (GEM). The classification is defined by the Hong Kong Stock Exchange. The table also report institutional holdings based on regions of headquarters. The region classification is defined by National Bureau of Statistics of China, by which the East region includes Beijing, Fujian, Guangdong, Hainan, Hebei, Jiangsu, Shandong, Shanghai, Tianjin, and Zhejiang. This study lists firms headquartered in Beijing and Guangdong separately, as the former are likely to subject to political influence and latter are culturally close to Hong Kong. The Middle region includes Anhui, Henan, Hubei, Hunan, Jiangxi, and Shanxi. The northeast region includes Heilongjiang, Jilin and Liaoning. The West region includes Chongqing, Gansu, Guangxi, Guizhou, Neimenggu (Inner Mongolia), Ningxia, Qinghai Shanxi, Sichuan, Xinjiang, Xizang (Tibet), and Yunnan. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The sample include 2,603 Chinese firms.

	All	Overseas	Beijing	Guangdong	East	Middle and Northeast	West
Whole sample	0.90	1.24	1.25	0.44	0.45	0.44	0.24
Domestic listed firms	0.36	--	0.82	0.32	0.32	0.16	0.16
Hong Kong-listed firms	2.28	1.24	4.14	2.83	1.15	2.82	2.10
H-shares	3.00	--	4.64	2.83	1.29	3.10	2.10
Red-chips	1.27	1.27	--	--	--	--	--
Mainboard	2.31	1.31	4.14	1.50	1.19	2.82	2.33
GEM	3.17	0.00	--	9.50	0.00	--	0.00
AH-listed firms	2.10	--	0.71	1.00	4.89	2.50	0.00

Table 8

Cross-sectional probit regressions: Firms from different regions listing on the Hong Kong Stock Exchange. This table report the probit regressions that estimate the probability that a Chinese firm from different regions listed on the Hong Kong Stock Exchange. The sample is divided into five subsamples based on the region classification is defined by National Bureau of Statistics of China, by which the East region includes Beijing, Fujian, Guangdong, Hainan, Hebei, Jiangsu, Shandong, Shanghai, Tianjin, and Zhejiang. Panel A and B report the results for firms headquartered in Beijing and Guangdong, respectively, the former are likely to subject to political influence and latter are culturally close to Hong Kong. Panel C report results for other part of the East region. Panel D reports the results for Middle and Northeast regions. The Middle region includes Anhui, Henan, Hubei, Hunan, Jiangxi, and Shanxi. The northeast region includes Heilongjiang, Jilin and Liaoning. Panel E reports results for the West region, which includes Chongqing, Gansu, Guangxi, Guizhou, Neimenggu (Inner Mongolia), Ningxia, Qinghai Shanxi, Sichuan, Xinjiang, Xizang (Tibet), and Yunnan. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. Institutional Holding is the percentage of strategic free float shares held by institutional investors. Institution Index is a dummy variable that equals one if a firm has strategic institutional shareholder(s). The definitions and data sources for control variables are detailed in Appendix 1. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. Pseudo- R^2 is a goodness-of-fit measure according to the difference between restricted and unrestricted likelihood functions. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Panel A: Beijing		Panel B: Guangdong		Panel C: East		Panel D: Middle and Northeast		Panel E: West	
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Institutional Holding	0.047 (1.10)		0.122 (1.48)		0.167*** (5.08)		0.303*** (4.12)		0.188*** (3.12)	
Institution Index		1.663*** (5.80)		1.735*** (2.85)		0.946*** (5.26)		0.874** (2.19)		0.627 (1.41)
ROA	0.066*** (2.61)	0.104*** (3.77)	0.055 (1.63)	0.058 (1.57)	-0.009 (-0.64)	-0.013 (-0.89)	0.001 (0.98)	0.002 (1.36)	0.196*** (3.75)	0.194*** (3.75)
Sales growth	0.000 (0.12)	0.002 (0.63)	-0.001 (-0.14)	-0.003 (-0.44)	0.003* (1.86)	0.002 (1.32)	0.015*** (2.68)	0.015*** (2.65)	0.007 (1.24)	0.005 (0.93)
Firm size	0.580*** (7.42)	0.647*** (5.78)	0.419** (2.51)	0.627*** (3.15)	0.202** (2.50)	0.223*** (2.74)	0.468*** (3.53)	0.551*** (4.02)	0.409*** (3.03)	0.443*** (3.38)
Leverage	-0.006 (-0.91)	-0.003 (-0.37)	-0.004 (-0.27)	-0.014 (-0.95)	-0.007 (-1.18)	-0.010* (-1.74)	0.001 (0.14)	0.001 (0.05)	0.01 (0.84)	0.008 (0.66)
Offer price	-0.097*** (-3.70)	-0.115*** (-4.21)	-0.248*** (-5.18)	-0.298*** (-5.44)	-0.059*** (-2.74)	-0.056*** (-2.66)	-0.102*** (-2.74)	-0.109** (-2.36)	-0.233*** (-3.00)	-0.248*** (-3.14)
Financial flexibility	0.031 (0.22)	-0.042 (-0.25)	0.412** (2.10)	0.418** (2.010)	0.344*** (3.48)	0.261*** (2.62)	0.328* (1.67)	0.23 (1.18)	0.503** (2.05)	0.492** (2.09)
Difference in HDI	--	--	--	--	-8.037 (-1.50)	-7.79 (-1.48)	8.314 (0.37)	6.665 (0.36)	1.076 (0.06)	-1.911 (-0.10)
Log foreign trade	--	--	--	--	-0.709*** (-3.63)	-0.615*** (-2.96)	-0.432 (-0.52)	-0.38 (-0.55)	1.036** (2.19)	1.068* (1.95)
Log GDP	--	--	--	--	0.685 (0.66)	0.489 (0.45)	2.680* (1.89)	3.147** (2.05)	-0.698 (-0.34)	-0.763 (-0.35)
Constant	-14.255*** (-2.92)	-15.514*** (-3.08)	-6.251* (-1.81)	-10.110** (-2.54)	3.85 (0.38)	3.523 (0.34)	-32.997 (-1.21)	-40.357 (-1.62)	-25.676 (-0.81)	-25.632 (-0.78)
Industry-dummy	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.567	0.658	0.668	0.714	0.363	0.353	0.618	0.584	0.599	0.582
N of Observation	338	338	289	289	750	750	212	212	181	116

Table 9

Cross-sectional multinomial logistic regressions: Foreign institutional investors and cross-border listing. This table reports coefficients on the strategic shareholdings of foreign institutional investors with a set of control variables regarding the types of cross-border listings of Chinese firms. The baseline group is domestically listed Chinese firms. Panel A report the result with foreign institutional investors' shareholding as the key variable of interest, which is the percentage of strategic free float shares held by foreign institutional investors. Panel B reports foreign institutional investors as shareholders, which is a dummy variable that equals one if a firm has strategic foreign institutional shareholder(s). The types of cross-border listings are classified by two ways. The first way is based on the channel through which a Chinese firm list to the Hong Kong Stock Exchange, including H-shares, Red-chips and AH-listings. The second way is based on the listing venues of the Hong Kong Stock Exchange, including the mainboard and the Growth Enterprise Market (GEM). The foreign institutional shareholding information is obtained from Orbis and for the period from 2000 to 2017. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. Data used for constructing control variables are obtained from Datastream. The definitions and data sources for control variables are detailed in Appendix 1. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. Pseudo- R^2 is a goodness-of-fit measure according to the difference between restricted and unrestricted likelihood functions. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	All Cross- border listing	H-share	Red-chips	AH-listing	Mainboard	GEM
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Foreign institutional investors' shareholding						
Institutional Holding	0.035*** (4.38)	0.067*** (3.36)	0.093** (2.27)	0.043* (1.85)	0.067*** (3.68)	0.487*** (23.12)
ROA	0.028 (1.14)	0.042 (0.66)	0.143 (1.63)	0.086 (1.38)	0.046 (0.73)	-0.760*** (-7.12)
Sales growth	0.004 (1.110)	0.008 (0.89)	-0.042** (-2.52)	0.018** (2.14)	0.001 (0.14)	0.307*** (20.62)
Firm size	0.397*** (5.11)	0.462*** (2.81)	0.648* (1.69)	1.219*** (4.68)	0.550*** (3.20)	-2.828*** (-6.09)
Leverage	-0.006 (-0.86)	0.006 (0.42)	0.004 (0.14)	-0.044** (-2.15)	0.005 (0.31)	-0.225*** (-8.13)
Offer price	-0.031*** (-2.88)	-0.038* (-1.67)	-0.373* (-1.80)	-0.078*** (-2.81)	-0.042* (-1.84)	-1.071*** (-15.25)
Financial flexibility	0.153 (1.26)	0.447* (1.70)	1.374 (1.10)	-0.414 (-0.84)	0.455* (1.71)	4.785*** (7.61)
Difference in HDI	-12.429* (-1.82)	-28.165* (-1.78)	9.714 (0.42)	-12.99 (-0.58)	-18.595 (-1.26)	-409.753*** (-13.66)
Log foreign trade	-0.065 (-0.54)	-0.354 (-1.42)	-0.697 (-0.87)	0.194 (0.44)	-0.287 (-1.28)	-4.928*** (-6.91)
Log GDP	-1.259 (-1.18)	-2.334 (-0.97)	6.412 (0.91)	-2.337 (-0.70)	-0.978 (-0.44)	-39.337*** (-8.41)
Constant	6.97 (0.58)	23.386 (0.85)	-75.035 (-1.05)	-7.022 (-0.19)	4.485 (0.17)	627.024*** (-12.69)
Industry-dummy	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.535	0.504	0.532	0.459	0.440	0.514
N of Observation	388	388	388	388	388	388
Panel B: The presence of foreign institutional shareholder(s)						
Institution Index	1.640*** (9.09)	3.083*** (8.37)	3.667*** (4.50)	1.952*** (3.80)	3.008*** (8.52)	80.564*** (28.28)
ROA	0.006 (0.33)	0.004 (0.07)	0.175*** (2.71)	0.110*** (3.77)	0.022 (0.19)	0.057 (1.02)
Sales growth	0.007*** (3.18)	0.013*** (2.96)	-0.002 (-0.10)	0.012* (1.92)	0.009** (2.03)	1.166*** (35.75)
Firm size	0.201*** (3.39)	0.086 (0.64)	0.584** (2.00)	1.010*** (5.80)	0.293* (1.74)	-62.004*** (-35.59)
Leverage	-0.003 (-0.60)	0.009 (0.730)	-0.003 (-0.19)	-0.033** (-2.42)	0.002 (0.16)	1.948*** (21.39)
Offer price	-0.070*** (-4.74)	-0.130** (-2.46)	-0.346*** (-4.81)	-0.130*** (-3.58)	-0.119** (-2.49)	-6.317*** (-23.94)
Financial flexibility	0.176* (1.66)	0.422* (1.68)	0.833** (2.08)	-0.142 (-0.45)	0.422 (1.52)	17.822*** (24.32)
Difference in HDI	-3.399 (-0.82)	-6.888 (-0.77)	26.14 (1.62)	-2.802 (-0.18)	-1.769 (-0.19)	-332.848*** (-6.25)
Log foreign trade	0.032 (0.36)	-0.097 (-0.49)	-0.338 (-0.85)	0.36 (1.16)	-0.129 (-0.66)	-29.946*** (-23.19)
Log GDP	0.224 (0.33)	1.016 (0.69)	7.556** (2.57)	-0.693 (-0.28)	1.793 (1.13)	103.086*** (9.74)
Constant	-7.918 (-1.08)	-12.491 (-0.79)	-98.611*** (-3.08)	-24.113 (-0.89)	-25.797 (-1.52)	727.248*** (7.36)
Industry-dummy	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.535	0.504	0.532	0.459	0.440	0.514
N of Observation	531	531	531	531	531	531

Table 10

The effect of institutional investors on cross-border listing: Government holding regressions. This table reports results from OLS regressions that estimate the effect of cross-border listings on the government ownership. The sample is constructed based on Orbis that indicates the global ultimate owner is the Chinese government. The dependent variable in each regression is the percentage of shares owned by the largest shareholders. The key variable of interest is institutional investors' shareholding, measured by Institutional Holding that is the percentage of strategic free float shares held by foreign institutional investors and by Institutional Index that is a dummy variable that equals one if a firm has strategic foreign institutional shareholder(s). H-share listing, Red-chips listing, AH-listing, and HK mainboard listing are dummy variables that equals one if a firm lists its shares via H-shares, Red-chips, AH-listings, or on the Hong Kong mainboard. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The definitions and data sources for control variables are detailed in Appendix 1. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. The t-statistics are reported in the parentheses where standard errors are clustered on firms. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable	Government Ownership							
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Institutional Holding	-1.077*** (-3.01)	-1.118** (-2.49)	-0.638* (-1.73)	-1.208*** (-2.73)				
Institutional Index					-5.288 (-1.49)	-4.789 (-1.17)	-1.279 (-0.24)	-5.395 (-1.21)
H-share listing		-8.728** (-2.02)				-9.506 (-1.50)		
Red-chips listing			9.402** (2.18)				10.067 (1.60)	
AH-listing				-2.634 (-0.57)				0.321 (0.06)
Institution* H-share		0.567 (0.92)				1.917 (0.25)		
Institution * Red-chips			0.065 (0.07)				2.935 (0.32)	
Institution * AH-listing				0.451 (0.73)				0.000 (0.00)
ROA	0.187** (2.10)	0.173* (1.88)	0.165* (1.74)	0.189** (2.13)	0.164* (1.92)	0.148* (1.67)	0.133 (1.41)	0.164* (1.91)
Sales growth	0.001 (0.84)	0.001 (0.54)	0.001 (0.46)	0.001 (0.80)	0.001 (0.71)	0.001 (0.43)	0.001 (0.53)	0.001 (0.70)
Firm size	2.309** (2.46)	2.541*** (2.75)	3.066*** (3.22)	2.418** (2.55)	2.381** (2.43)	2.642*** (2.73)	3.106*** (3.27)	2.370** (2.51)
Leverage	-0.008** (-2.13)	-0.008* (-1.92)	-0.008** (-2.05)	-0.008** (-2.18)	-0.008** (-2.11)	-0.007* (-1.93)	-0.007* (-1.89)	-0.008** (-2.11)
Offer price	-0.092 (-0.15)	-0.187 (-0.31)	-0.204 (-0.34)	-0.103 (-0.17)	-0.018 (-0.03)	-0.136 (-0.24)	-0.153 (-0.27)	-0.019 (-0.03)
Financial flexibility	0.342 (0.20)	0.262 (0.16)	0.168 (0.10)	0.326 (0.19)	0.294 (0.16)	0.295 (0.17)	0.134 (0.08)	0.301 (0.17)
Constant	5.766 (0.35)	2.287 (0.14)	-13.179 (-0.73)	4.642 (0.28)	5.817 (0.35)	1.65 (0.10)	-14.02 (-0.75)	5.934 (0.36)
Industry-dummy	YES	YES	YES	YES	YES	YES	YES	YES
Pseudo- R^2	0.107	0.124	0.131	0.106	0.089	0.112	0.123	0.088
N of Observation	934	934	934	934	934	934	934	934

Table 11

The effect of institutional investors on cross-border listing: Analyst coverage regressions. This table reports results from OLS regressions that estimate the effect of cross-border listings on analyst coverage. The dependent variable in each regression is the number of analysts. The key variable of interest is institutional investors' shareholding, measured by Institutional Holding that is the percentage of strategic free float shares held by foreign institutional investors and by Institutional Index that is a dummy variable that equals one if a firm has strategic foreign institutional shareholder(s). H-share listing, Red-chips listing, AH-listing, mainboard listing, and A-share listing are dummy variables that equals one if a firm lists its shares via H-shares, Red-chips, AH-shares, on the Hong Kong mainboard, and in Chinese domestic exchanges, respectively. Earning surprise is the annual absolute value of difference between actual and forecasted earning per shares, scaled by the mean of the forecasted earning per shares. Forecast dispersion is the annual standard deviation of the forecasted earning per shares, scaled by the mean of the forecasted earning per shares. Firm size is the natural logarithm of total assets. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. The definitions and data sources for variables are detailed in Appendix 1. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable	Analyst Coverage					
Model	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Institutional investors' shareholding						
Institutional Holding	0.164*** (2.78)	0.156** (2.51)	0.170*** (2.88)	0.116** (1.99)	0.134** (2.28)	0.029 (0.54)
H-share listing		0.284 (0.32)				
Red-chip listing			1.714** (2.09)			
AH-share listing				2.971*** (4.20)		
Mainboard listing					1.236** (2.01)	
A-share listing						-3.327*** (-6.33)
Earning surprise	0.493 (1.05)	0.494 (1.05)	0.228 (0.53)	0.663 (1.36)	0.307 (0.69)	0.182 (0.41)
Forecast dispersion	-1.919*** (-3.34)	-1.919*** (-3.33)	-1.622*** (-2.96)	-2.061*** (-3.48)	-1.704*** (-3.05)	-1.500*** (-2.70)
Firm size	3.067*** (23.20)	3.064*** (23.01)	3.034*** (23.39)	2.777*** (18.24)	3.029*** (23.13)	2.640*** (19.98)
Constant	-40.819*** (-19.82)	-40.779*** (-19.76)	-40.503*** (-20.11)	-37.240*** (-15.86)	-40.415*** (-20.04)	-32.395*** (-14.27)
Industry dummy	YES	YES	YES	YES	YES	YES
Adjusted R ²	0.5196	0.5196	0.5234	0.5349	0.5226	0.5453
N of Observation	5,368	5,368	5,368	5,368	5,368	5,368
Panel B: The presence of institutional shareholder(s)						
Institutional Holding	1.361*** (3.76)	1.305*** (3.64)	1.359*** (3.80)	1.031*** (2.83)	1.228*** (3.38)	0.672* (1.90)
H-share listing		0.551 (0.67)				
Red-chip listing			1.615** (1.97)			
AH-share listing				2.902*** (4.20)		
Mainboard listing					1.287** (2.13)	
A-share listing						-3.182*** (-6.39)
Earning surprise	0.497 (1.07)	0.501 (1.08)	0.245 (0.57)	0.664 (1.37)	0.306 (0.69)	0.209 (0.47)
Forecast dispersion	-1.902*** (-3.31)	-1.904*** (-3.31)	-1.621*** (-2.95)	-2.046*** (-3.46)	-1.683*** (-3.02)	-1.519*** (-2.73)
Firm size	3.051*** (23.03)	3.042*** (22.65)	3.022*** (23.42)	2.769*** (17.83)	3.007*** (22.97)	2.633*** (19.62)
Constant	-41.044*** (-19.89)	-40.924*** (-19.76)	-40.768*** (-20.24)	-37.466*** (-15.84)	-40.545*** (-20.15)	-32.706*** (-14.59)
Industry dummy	YES	YES	YES	YES	YES	YES
Adjusted R ²	0.5221	0.5223	0.5255	0.5367	0.5256	0.5469
N of Observation	5,368	5,368	5,368	5,368	5,368	5,368

Table 12

Cox model predicting institutional investors as strategic holdings. This table reports the results from Cox model that estimate the determinants of institutional holdings using the ex-post analysis, given that the firm does not have strategic institutional investors at the time of listing. ROA is return on assets. Sales growth is the annual percentage change in sales. Firm size is the natural logarithm of total assets. Leverage is the ratio of total debt to total capital. Offer price is the stock price initial offer price in Hong Kong dollars. Financial flexibility is an index ranging from zero to three, where values increase by one point if the firm is at the position above the 75th percentiles of cash ratio, by one point if the firm is at the position above the 75th percentiles of dividend ratio, and by one point if the firm is at the position below the 25th percentiles of the capital expenditure to total assets ratio. Difference in HDI is the difference in the human development index between Hong Kong and the province for a given firm. Log Foreign Trade is the natural logarithm of the foreign trading volume of the province for a given firm. Log GDP is the natural logarithm of GDP per capita of the province for a given firm. Analyst coverage is the annual count of analyst associated with a firm after listing. Earning surprise is the annual absolute value of difference between actual and forecasted earning per shares, scaled by the mean of the forecasted earning per shares. Forecast dispersion is the annual standard deviation of the forecasted earning per shares, scaled by the mean of the forecasted earning per shares. The detailed definitions and data sources for the regression variables are summarized in Appendix 1. The institutional shareholding information and data used for constructing control variables are obtained from Datastream. Data sources for information on listings are from the Hong Kong stock exchange, Shanghai Stock Exchange, and Bloomberg. The t-statistics are reported in the parentheses where standard errors are clustered at the firm level. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Whole sample	China incorporated firms	China incorporated cross-border listing
	(1)	(2)	(3)
ROA	0.001 (0.09)	0.001 (0.08)	-0.018 (-1.04)
Sales growth	0.002 (1.52)	0.002 (1.60)	0.002 (0.71)
Firm size	-0.193*** (-2.74)	-0.211*** (-2.97)	-0.127 (-0.90)
Leverage	0.005 (1.22)	0.005 (1.15)	-0.005 (-0.82)
Offer price	-0.015 (-1.39)	-0.015 (-1.38)	0.000 (-0.00)
Financial flexibility	0.188* (1.76)	0.196* (1.83)	0.152 (0.93)
Log foreign trade	0.013 (0.17)	-0.005 (-0.06)	0.148 (0.91)
Log GDP	-0.961* (-1.77)	-0.834 (-1.53)	-2.900** (-2.34)
Difference in HDI	-10.147*** (-2.92)	-10.008*** (-2.86)	-18.009** (-2.30)
Analyst coverage	0.053*** (3.49)	0.061*** (3.96)	0.011 (0.50)
Earning surprise	-0.272 (-1.01)	-0.346 (-1.28)	-0.173 (-0.50)
Forecast dispersion	0.274 (0.73)	0.466 (1.28)	-0.919 (-1.63)
Industry dummy	YES	YES	YES
Wald > χ^2	10,491.11***	9172.27***	59525.95***
N of Observation	3,698	3,637	555

Appendix 1: Variable description and data source

Variable	Description	Source
Institutional Holding	The percentage of strategic free float shares held by foreign institutional investors.	Datastream
Institutional Index	The dummy variable that equals one if a firm has strategic foreign institutional shareholder(s).	Datastream
ROA	The ratio of total income to total assets.	Datastream
Sales growth	The percentage of annual change in sales revenue.	Datastream
Firm size	The natural logarithm of total assets.	Datastream
Leverage	The total debt divided by the total capital of the firm	Datastream
Offer price	The stock price initial offer price in Hong Kong dollars.	Bloomberg
Offer size	The total amount of fund raised through public offering in Hong Kong dollars	Bloomberg
Financial flexibility	A count variable by adding one point for a firm with high cash and liquid assets, one point for high dividends, and one more point for low capital expenditures. The high and low value depends on whether cash and liquid asset holdings are greater than the 75th percentile, whether dividends are greater than the 75th percentile, and whether capital expenditures below the 25th percentile among sample firms.	Datastream
Log foreign trade	The foreign trading volume of the province for a given firm.	Ministry of commerce, people's republic of china (http://english.mofcom.gov.cn/column/statisticsie.shtml)
Log GDP	The log of GDP per capita of the province for a given firm.	National Bureau of Statistics of China (http://data.stats.gov.cn/english)
Differences in HDI	The difference in the human development index between Hong Kong and the province for a given firm.	the China National Human Development Report
Analyst coverage	The number of analysts	Datastream
Earning surprise	The annual absolute value of difference between actual and forecasted earning per shares, scaled by the mean of the forecasted earning per shares.	Datastream
Forecast dispersion	The annual standard deviation of the forecasted earning per shares, scaled by the mean of the forecasted earning per shares	Datastream
Government Ownership	The percentage of shares owned by the largest shareholders, where the global ultimate owner is the Chinese government.	Orbis